CORE Econ Micro

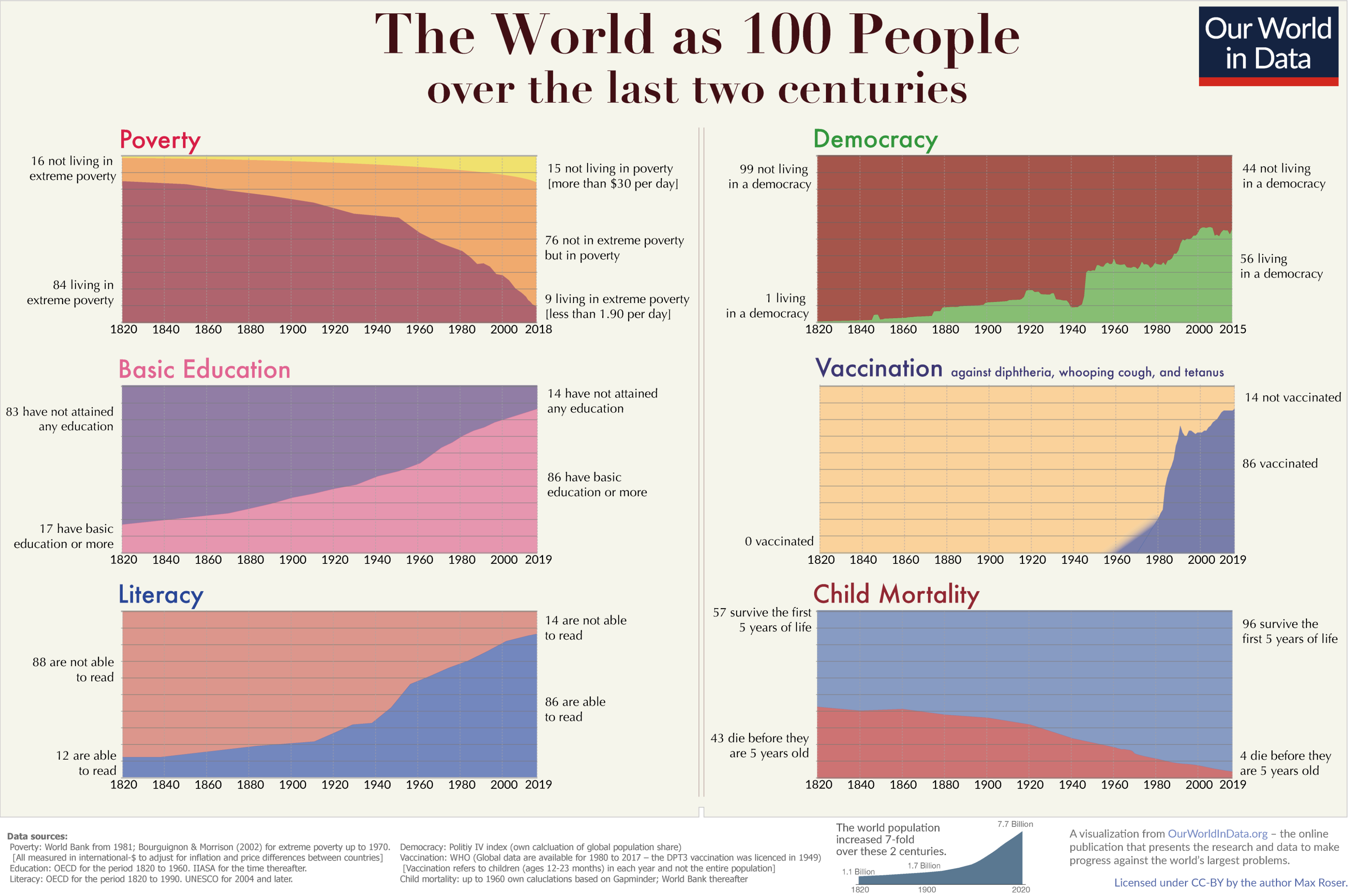
Prosperity, inequality, and planetary limits

Guillermo Woo-Mora

Paris Sciences et Lettres

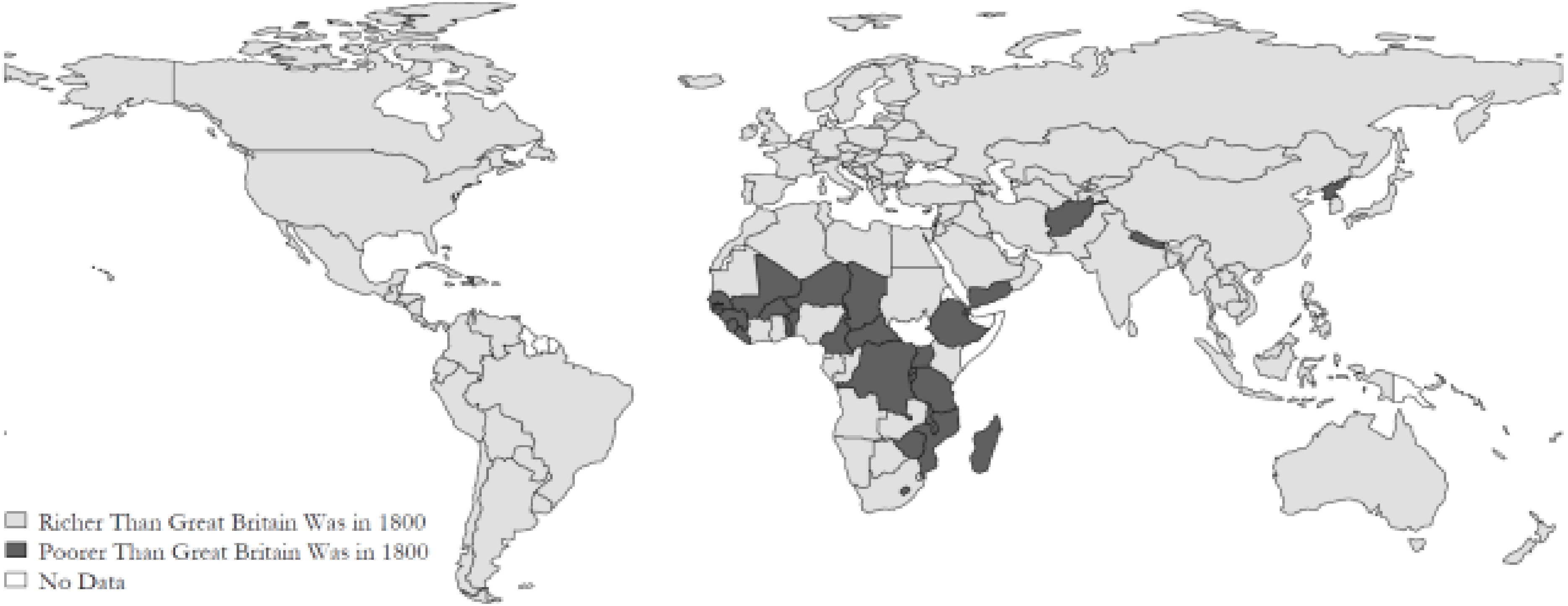
Autumn 2025

Are we better off?


The World as 100 People
over the last two centuries
Our World in Data
Poverty
16 not living in extreme poverty
84 living in extreme poverty
15 not living in poverty [more than $30 per day]
76 not in extreme poverty but in poverty
9 living in extreme poverty [less than 1.90 per day]
1820 1840 1860 1880 1900 1920 1940 1960 1980 2000 2018
Democracy
99 not living in a democracy
1 living in a democracy
44 not living in a democracy
56 living in a democracy
1820 1840 1860 1880 1900 1920 1940 1960 1980 2000 2015
Basic Education
83 have not attained any education
17 have basic education or more
14 have not attained any education
86 have basic education or more
1820 1840 1860 1880 1900 1920 1940 1960 1980 2000 2019
Vaccination against diphtheria, whooping cough, and tetanus
0 vaccinated
14 not vaccinated
86 vaccinated
1820 1840 1860 1880 1900 1920 1940 1960 1980 2000 2019
Literacy
88 are not able to read
12 are able to read
14 are not able to read
86 are able to read
1820 1840 1860 1880 1900 1920 1940 1960 1980 2000 2019
Child Mortality
57 survive the first 5 years of life
43 die before they are 5 years old
96 survive the first 5 years of life
4 die before they are 5 years old
1820 1840 1860 1880 1900 1920 1940 1960 1980 2000 2019
Data sources:
Poverty: World Bank from 1981; Bourguignon & Morrison (2002) for extreme poverty up to 1970.
[All measured in international-$ to adjust for inflation and price differences between countries]
Education: OECD for the period 1820 to 1960. IIASA for the time thereafter.
Literacy: OECD for the period 1820 to 1990. UNESCO for 2004 and later.
Democracy: Politiy IV index (own calcluation of global population share)
Vaccination: WHO (Global data are available for 1980 to 2017 – the DPT3 vaccination was licenced in 1949)
[Vaccination refers to children (ages 12-23 months) in each year and not the entire population]
Child mortality: up to 1960 own caluclations based on Gapminder; World Bank thereafter
The world population increased 7-fold over these 2 centuries.
7.7 Billion
1.1 Billion
1.7 Billion
1820
1900
2020
A visualization from OurWorldInData.org – the online publication that presents the research and data to make progress against the world's largest problems.
Licensed under CC-BY by the author Max Roser.

Countries Better-Off than GB in 1800

Koyama and Rubin, 2022



Countries Better-Off than US in 1900

Koyama and Rubin, 2022

Why are we better off?

Unit 1: Prosperity, inequality, and planetary limits

Unit 1: Prosperity, inequality, and planetary limits

- Economic inequality and divergence
- The technological revolution and growth
- The role of capitalism in economic growth
- Importance of the government in capitalist economies

> **The causes of the wealth and poverty of nations [were] the grand object of all enquiries in Political Economy** (Thomas Malthus to David Ricardo)
>
> **An Inquiry into the Nature and Causes of the Wealth of Nations** (Adam Smith)

Measuring income and living standards

Measuring income and living standards

What is a good measure of our well-being? For practical reasons, economist use (mainly) two:

Gross Domestic Product (GDP)

A measure of the market value of the output of final goods and services in the economy in a given period. Output of intermediate goods that are inputs to final production is excluded to prevent double counting.

GDP per capita

The average income of people in a country.

Disposable Income

$Total\ income – taxes + government\ transfers$

GDP per capita $\neq$ Disposable Income

Other measures of well-being?

- Human Development Index (Life expectancy, Education, Income)
- Subjective well-being (Life satisfaction, Happiness, Leisure)
- Nightlights

Comparing income at different times, and across different countries

- The starting point: **Nominal GDP**

$$Nominal\ GDP_y = \sum_i p_{iy} q_{iy}$$

- Taking account of price changes over time: **Real GDP**

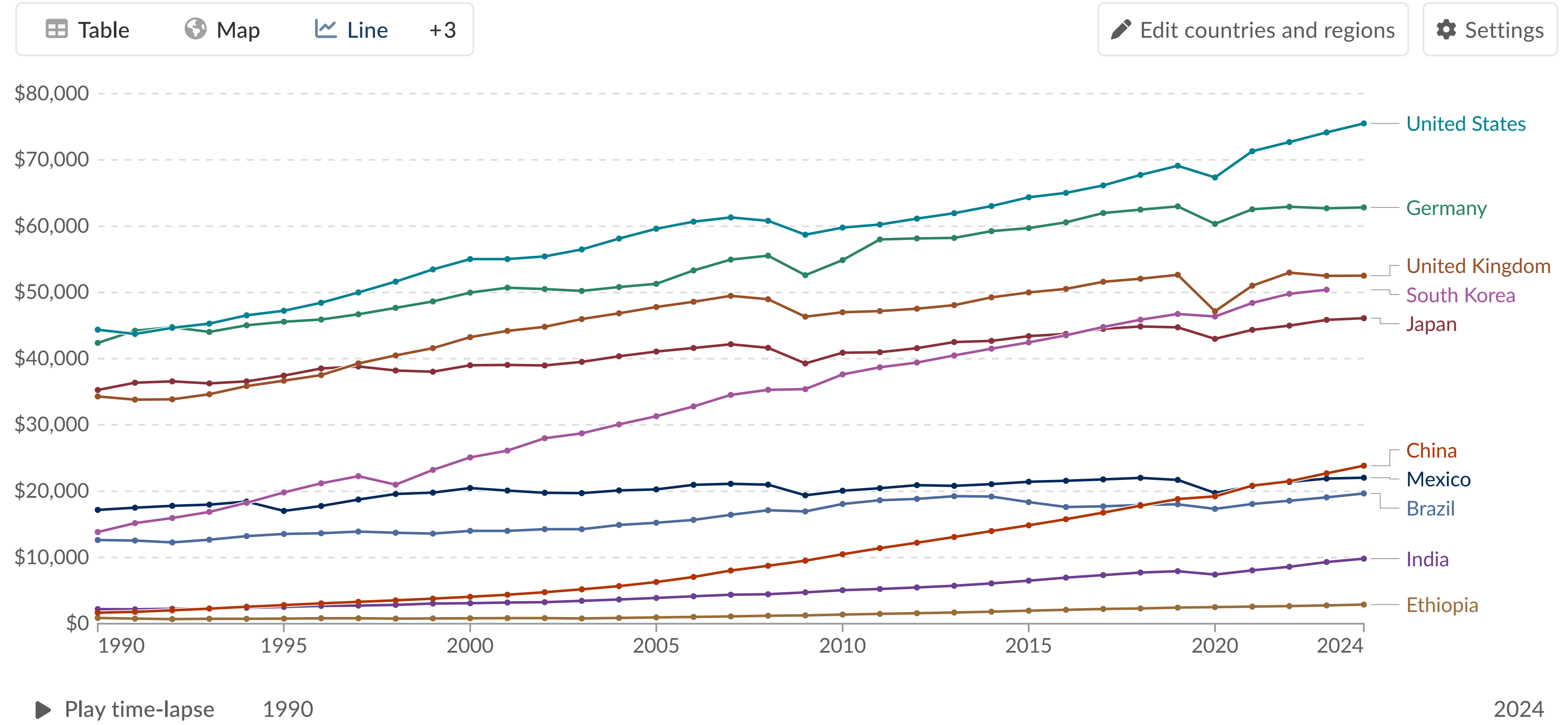
$$Real\ GDP_{y,\bar{y}} = \sum_i p_{i\bar{y}} q_{iy}$$

- Taking account of price differences among countries: **International prices and purchasing power**

Purchasing Power Parity (PPP) prices. Achieve parity (equality) in the real purchasing power.

GDP per capita

GDP per capita is a country's gross domestic product divided by its population. This data is adjusted for inflation and differences in living costs between countries.



Data source: Eurostat, OECD, IMF, and World Bank (2025) – Learn more about this data
Note: This data is expressed in international-$ at 2021 prices.

Explore the data →

History's hockey stick: Growth in income

Growth

Growth take-off occurred at different points in time for different countries:

- Britain was the first country to experience sustained economic growth. It began around 1650.
- In Japan, it occurred around 1870.
- The kink for China and India happened in the second half of the 20th century.

In some economies, substantial improvements in people's living standards did not occur until they gained independence from colonial rule or interference by European nations.

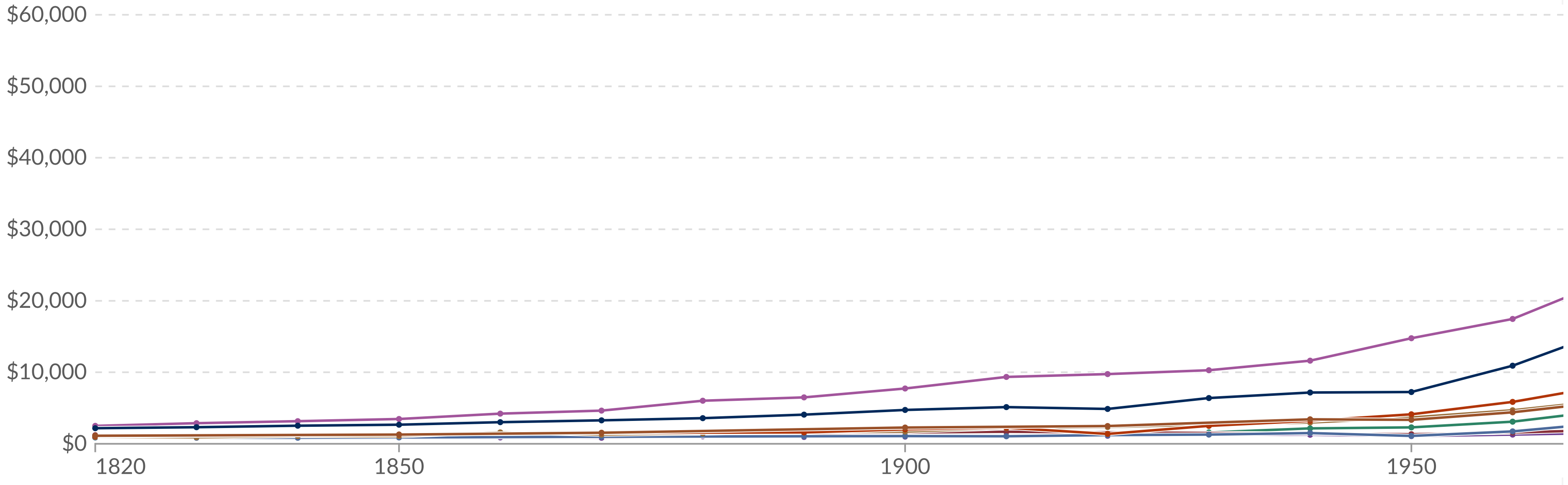
Hitorical data: The Maddison Project

GDP per capita, 1820 to 2022

Our World in Data

GDP per capita is a country's gross domestic product divided by its population. This data is adjusted for inflation and differences in living costs between countries.

Table Map Line +2 Edit countries and regions Settings



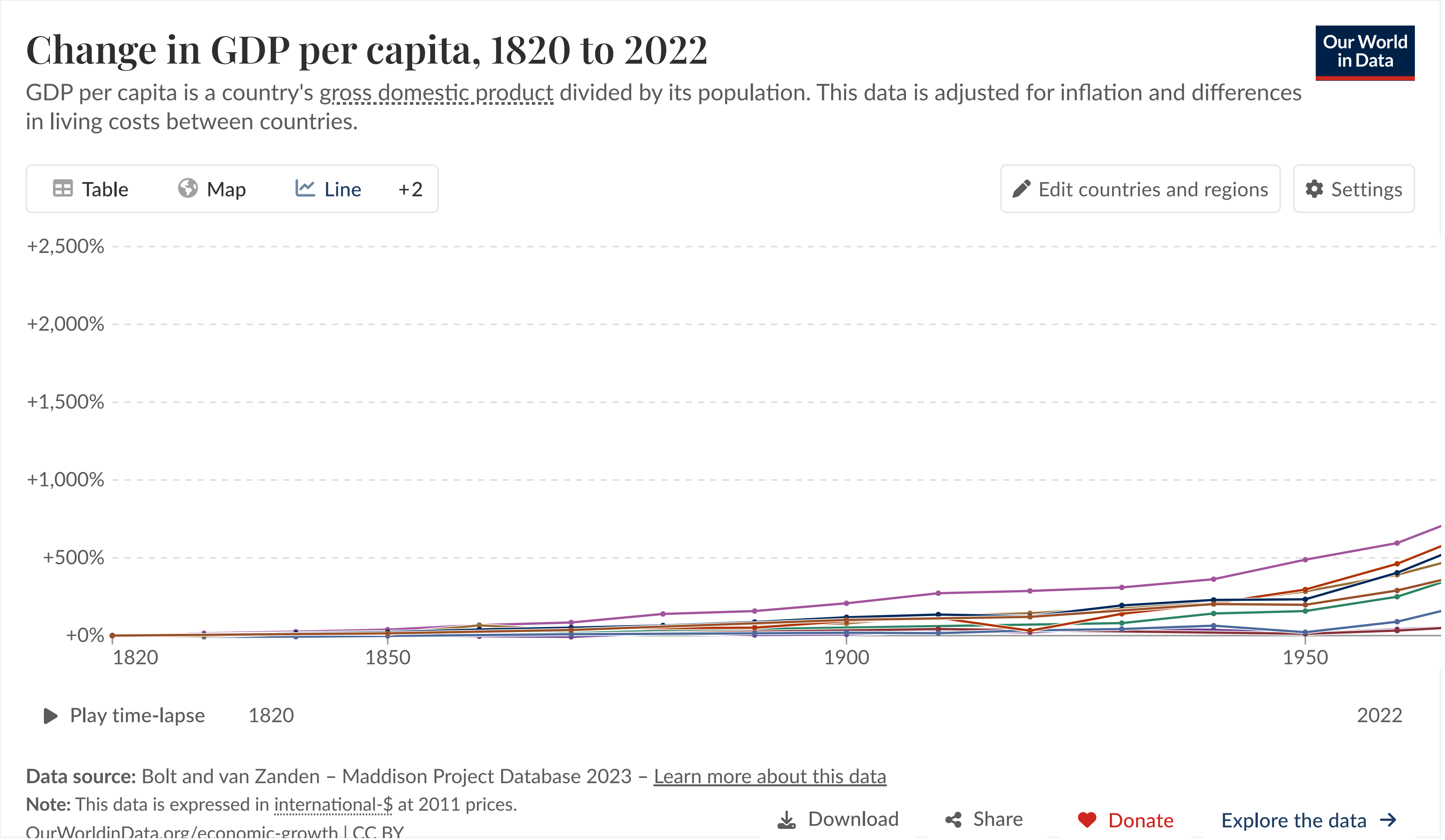
Play time-lapse 1820 2022

Data source: Bolt and van Zanden – Maddison Project Database 2023 – Learn more about this data

Note: This data is expressed in international-$ at 2011 prices.

Download Share Donate Explore the data →

$$growth\ rate = (y_{i1} - y_{i0})/y_{i0} = change\ in\ y/y\text{'s}\ original\ level$$



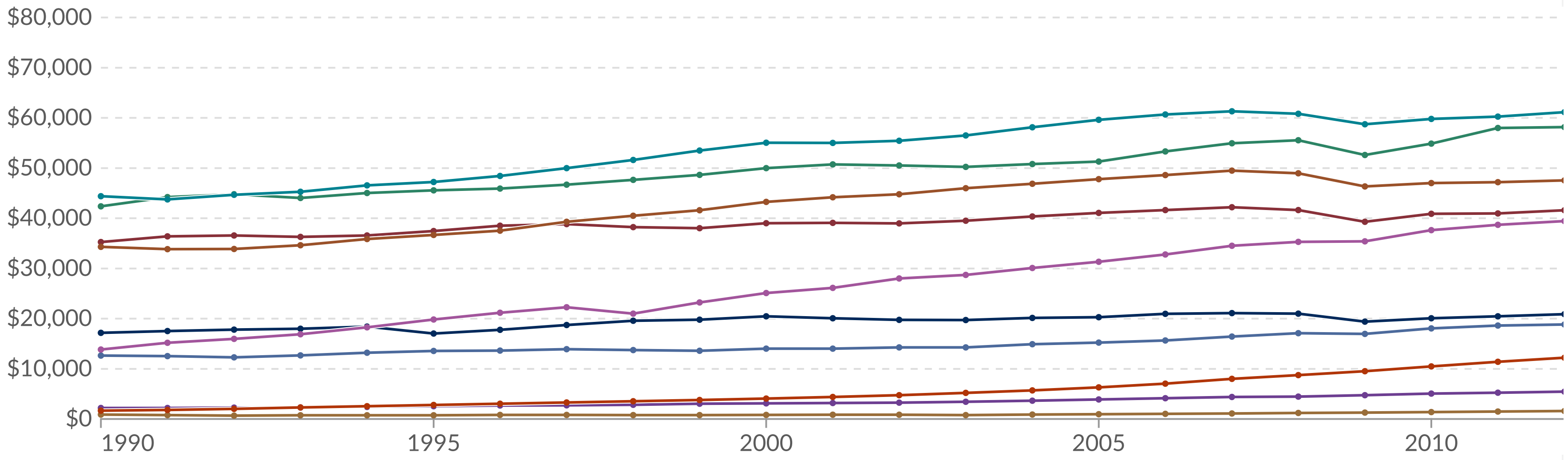
GDP Levels ≠ GDP Growth

GDP per capita

Our World in Data

GDP per capita is a country's gross domestic product divided by its population. This data is adjusted for inflation and differences in living costs between countries.

Table Map Line +3 Edit countries and regions Settings

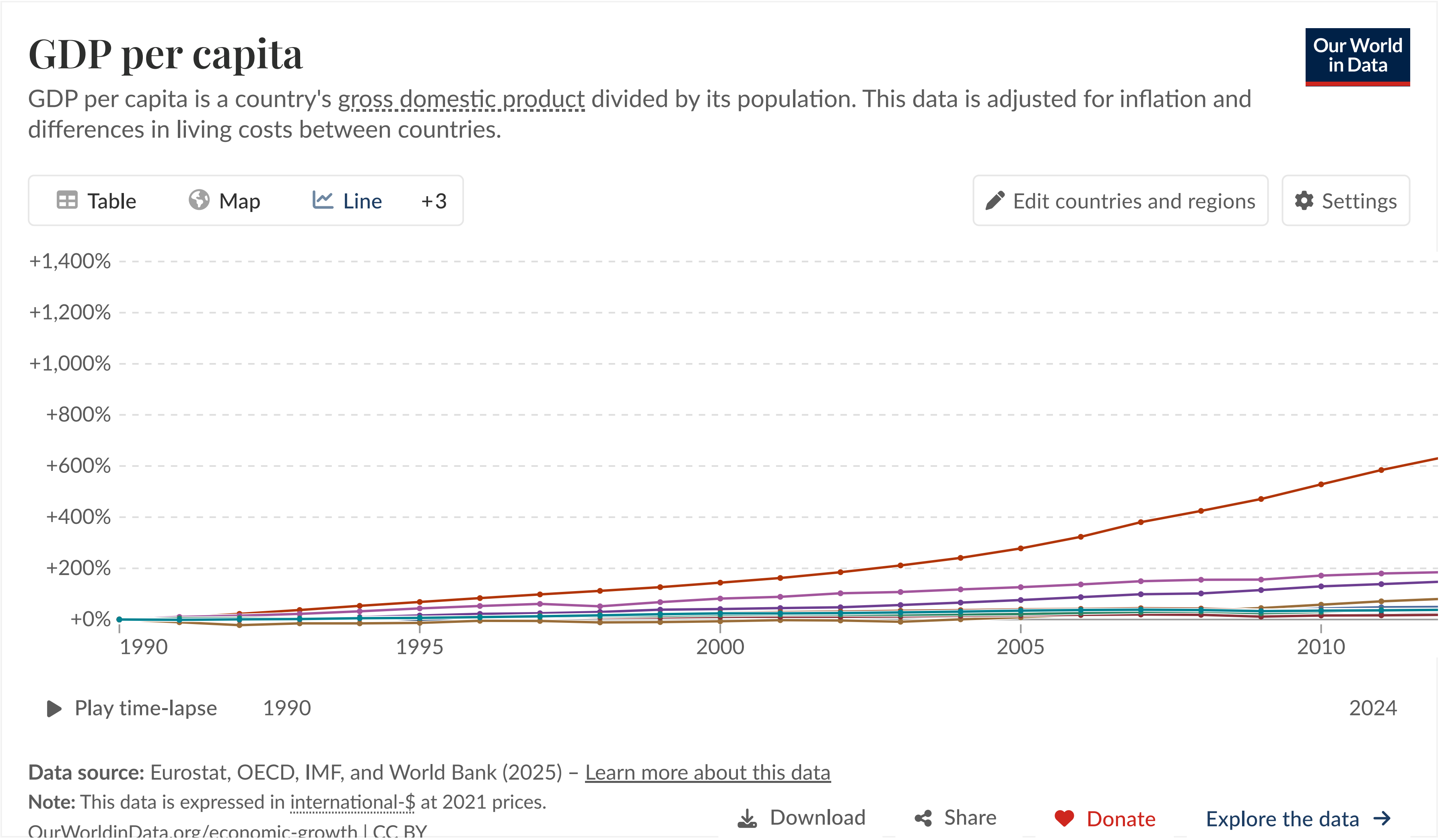


Play time-lapse 1990 2024

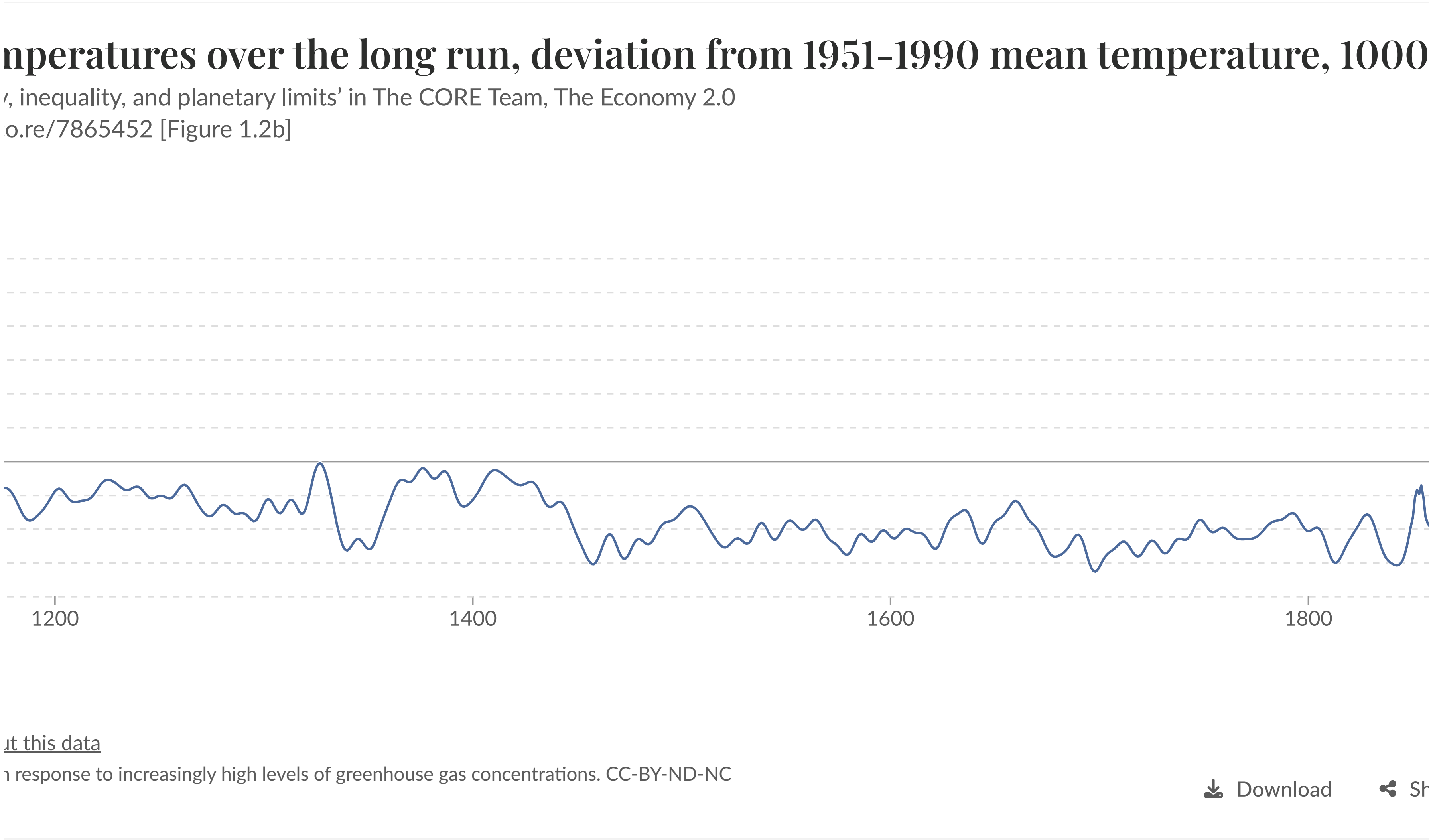
Data source: Eurostat, OECD, IMF, and World Bank (2025) – Learn more about this data
Note: This data is expressed in international-$ at 2021 prices.
OurWorldinData.org/economic-growth | CC BY

Download Share Donate Explore the data →

GDP Levels ≠ GDP Growth

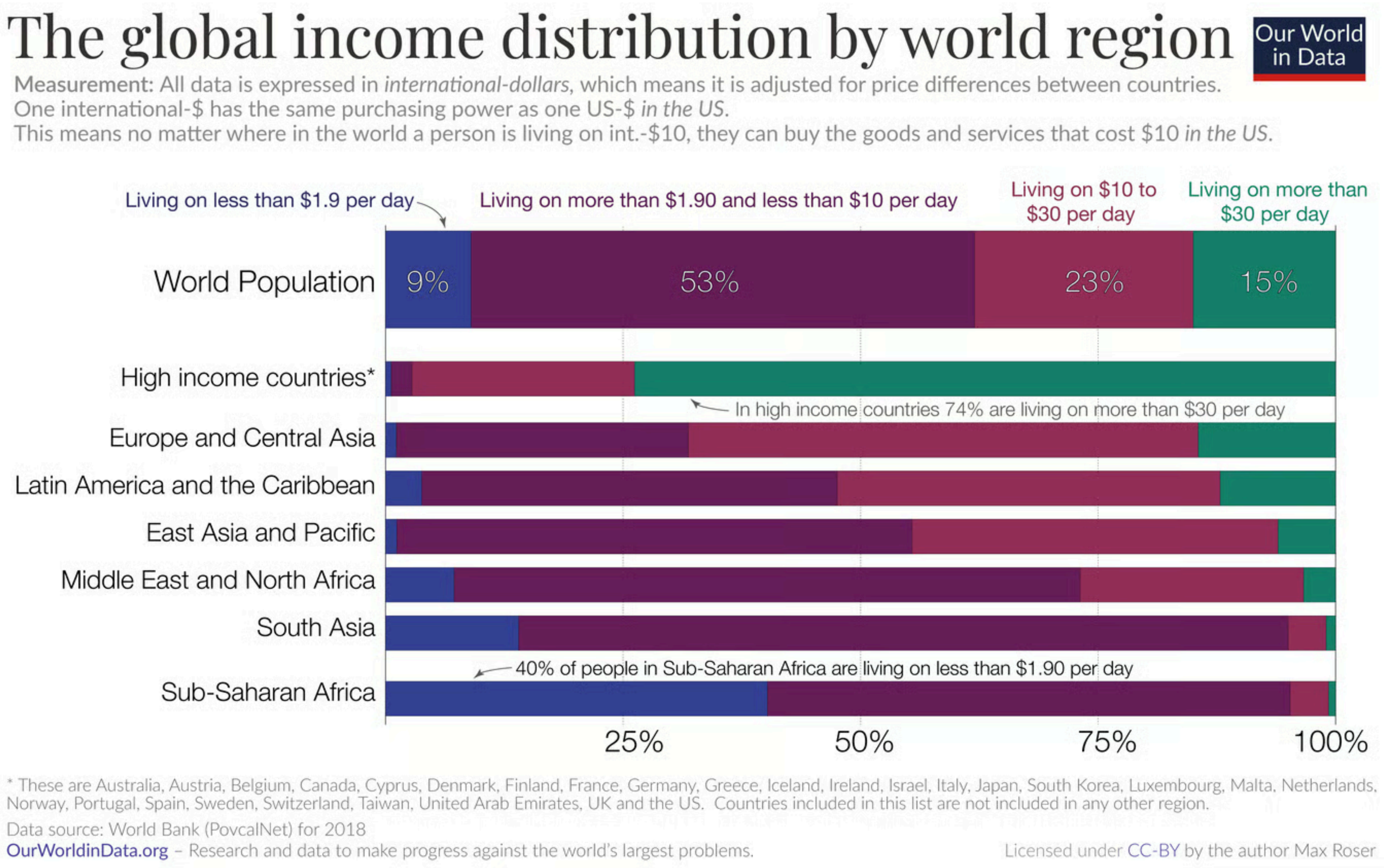


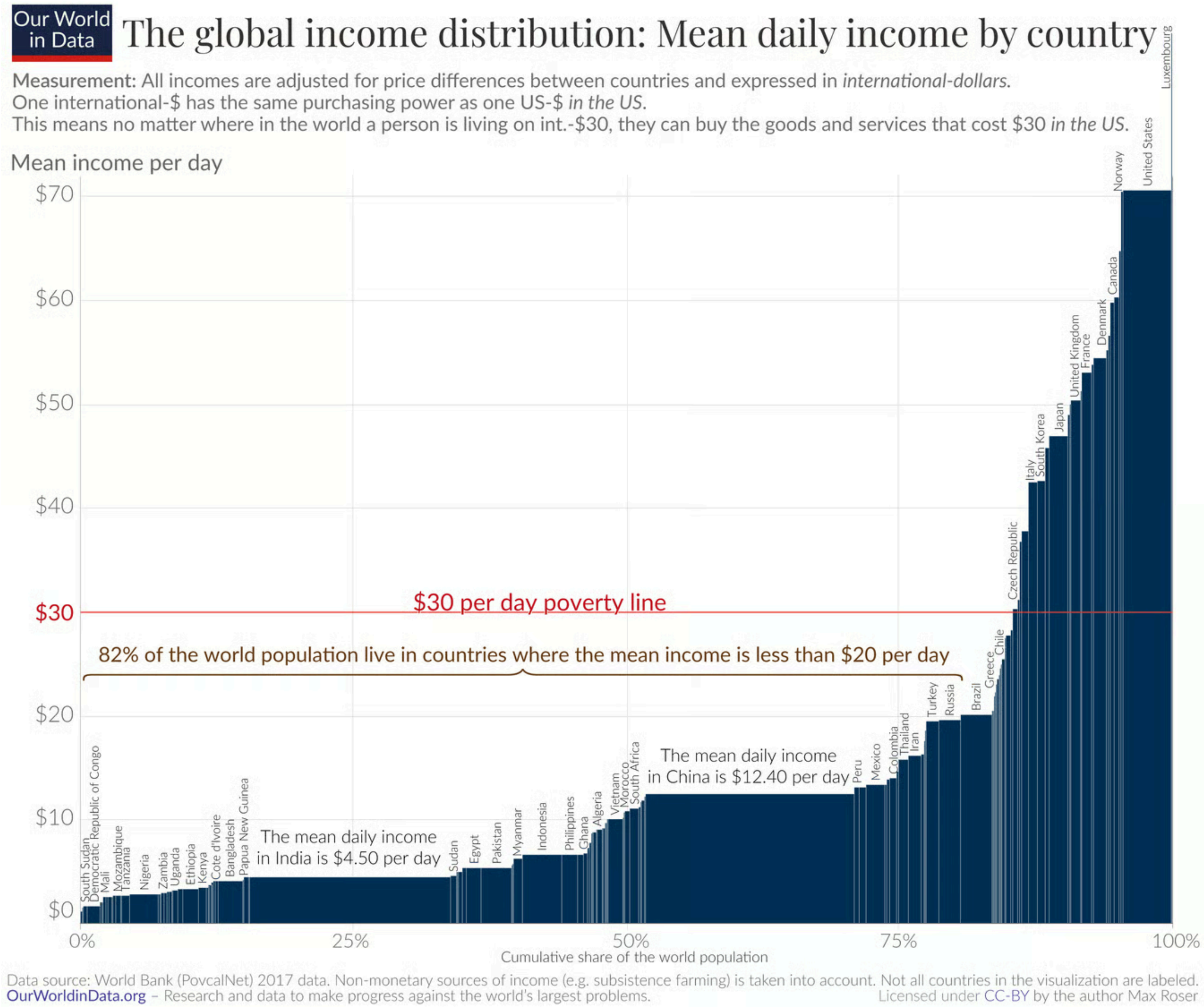
Another Hockey Stick: Climate change



Income Inequality

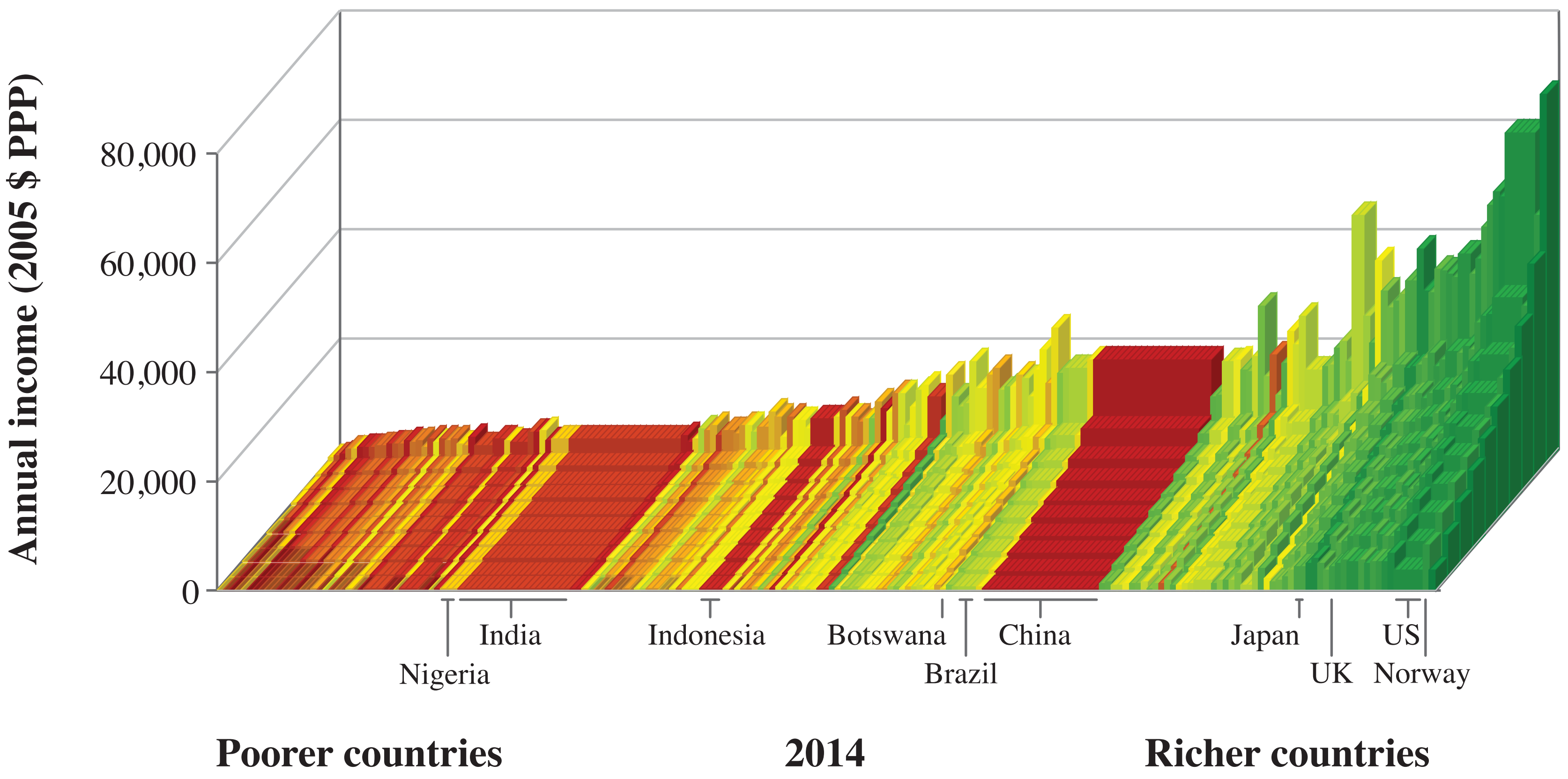
Global income distribution



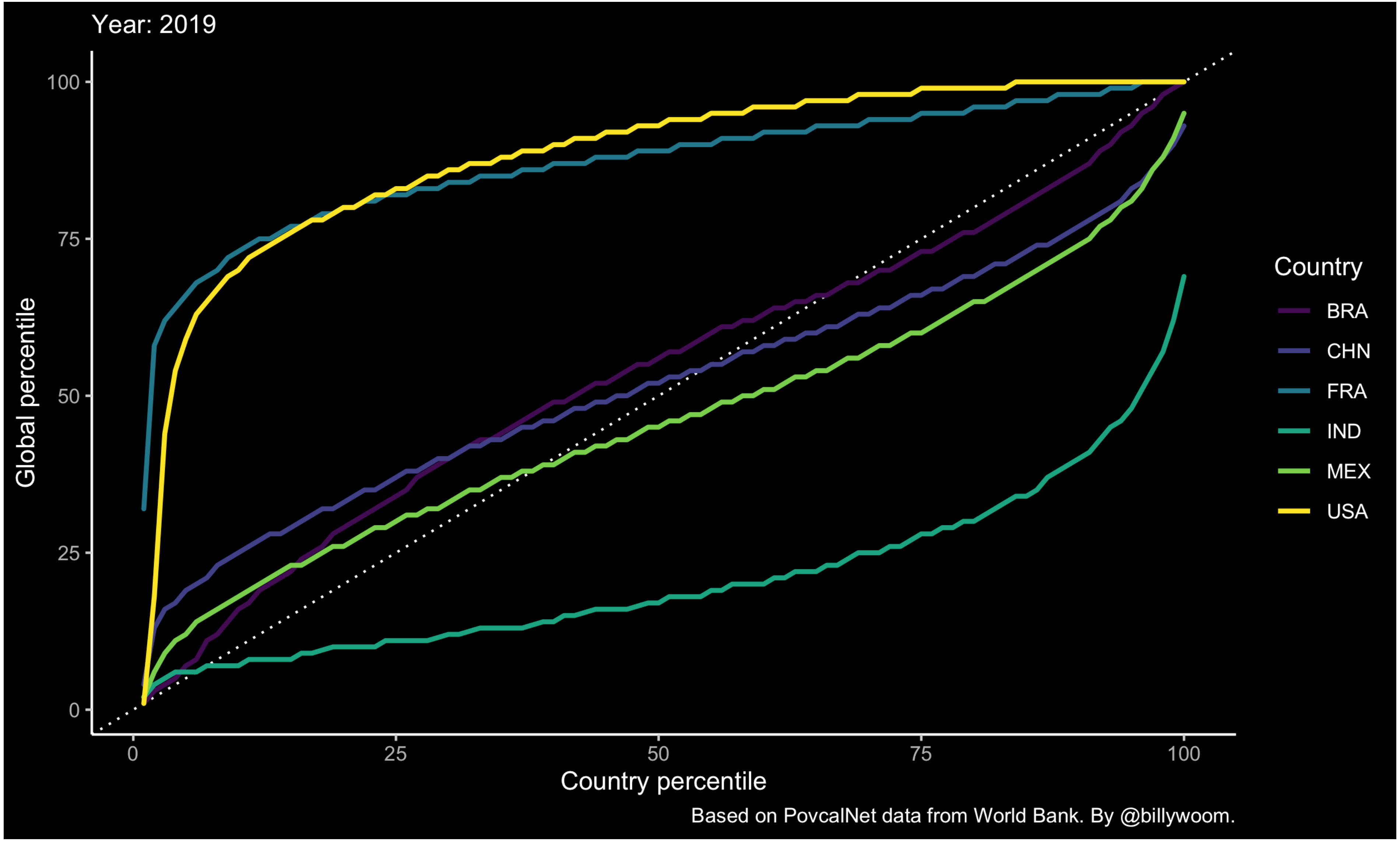


Our World in Data
The global income distribution: Mean daily income by country
Measurement: All incomes are adjusted for price differences between countries and expressed in international-dollars.
One international-$ has the same purchasing power as one US-$ in the US.
This means no matter where in the world a person is living on int.-$30, they can buy the goods and services that cost $30 in the US.
Mean income per day
$70
$60
$50
$40
$30
$20
$10
$0
$30 per day poverty line
82% of the world population live in countries where the mean income is less than $20 per day
The mean daily income in India is $4.50 per day
The mean daily income in China is $12.40 per day
South Sudan
Democratic Republic of Congo
Mali
Mozambique
Tanzania
Nigeria
Zambia
Uganda
Ethiopia
Kenya
Cote d'Ivoire
Bangladesh
Papua New Guinea
Sudan
Egypt
Pakistan
Myanmar
Indonesia
Philippines
Ghana
Algeria
Vietnam
Morocco
South Africa
Peru
Mexico
Colombia
Thailand
Iran
Turkey
Russia
Brazil
Greece
Chile
Czech Republic
Italy
South Korea
Japan
United Kingdom
France
Denmark
Canada
Norway
United States
Luxembourg
0%
25%
50%
75%
100%
Cumulative share of the world population
Data source: World Bank (PovcalNet) 2017 data. Non-monetary sources of income (e.g. subsistence farming) is taken into account. Not all countries in the visualization are labeled.
OurWorldinData.org – Research and data to make progress against the world's largest problems.
Licensed under CC-BY by the author Max Roser

Inequalities



- Between countries
- Within countries


Year: 2019
Global percentile
100
75
50
25
0
Country percentile
0
25
50
75
100
Country
BRA
CHN
FRA
IND
MEX
USA
Based on PovcalNet data from World Bank. By @billywoom.

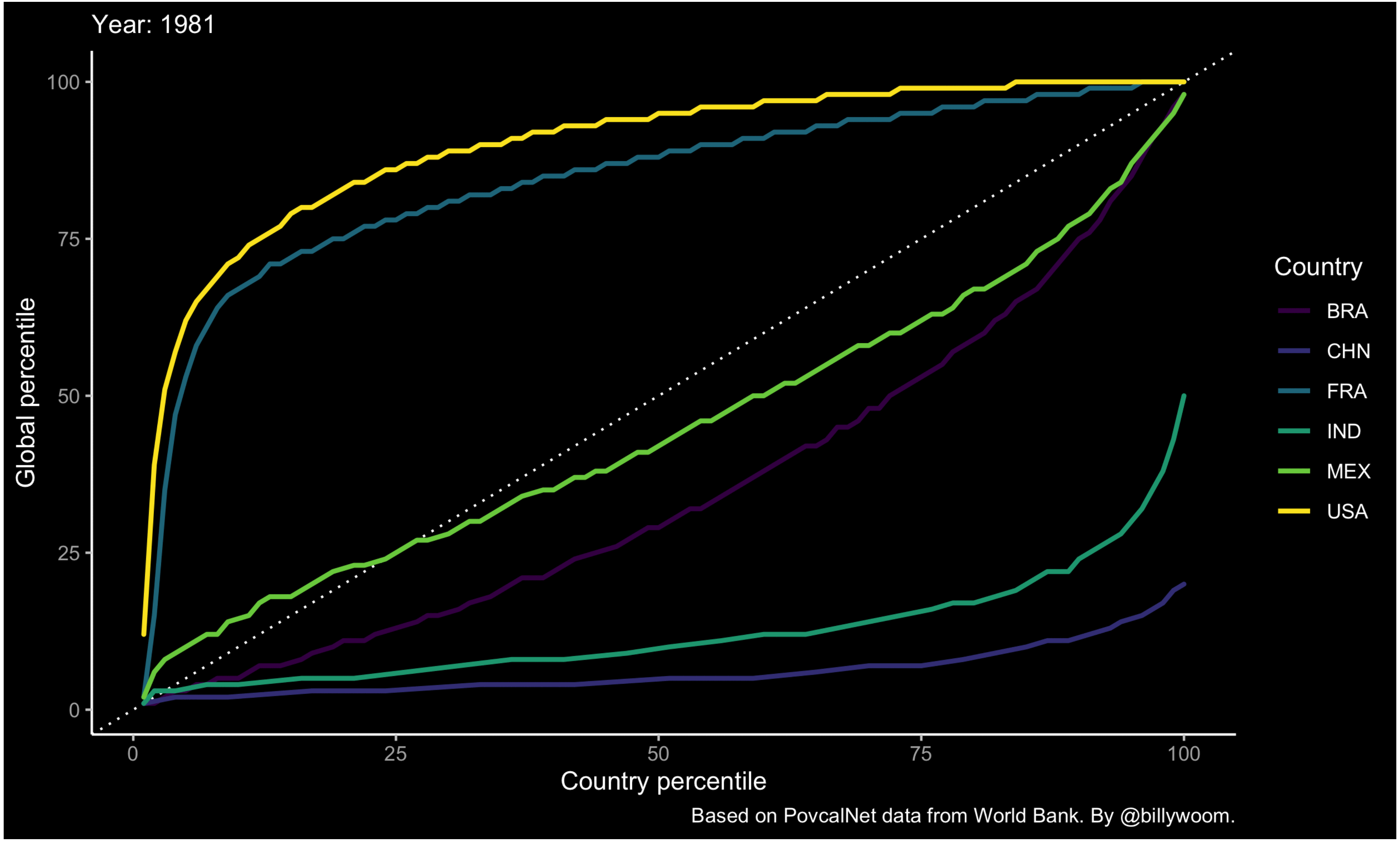
Inequality and Growth

Between-country inequalities:

- Countries that took off economically a century or more ago —UK, Japan, Italy— are now rich.
- The countries that took off only recently, or not at all, are in the flatlands.

Within-country inequalities:

- Does growth lift all boats?
- Inequality hinders economic development?

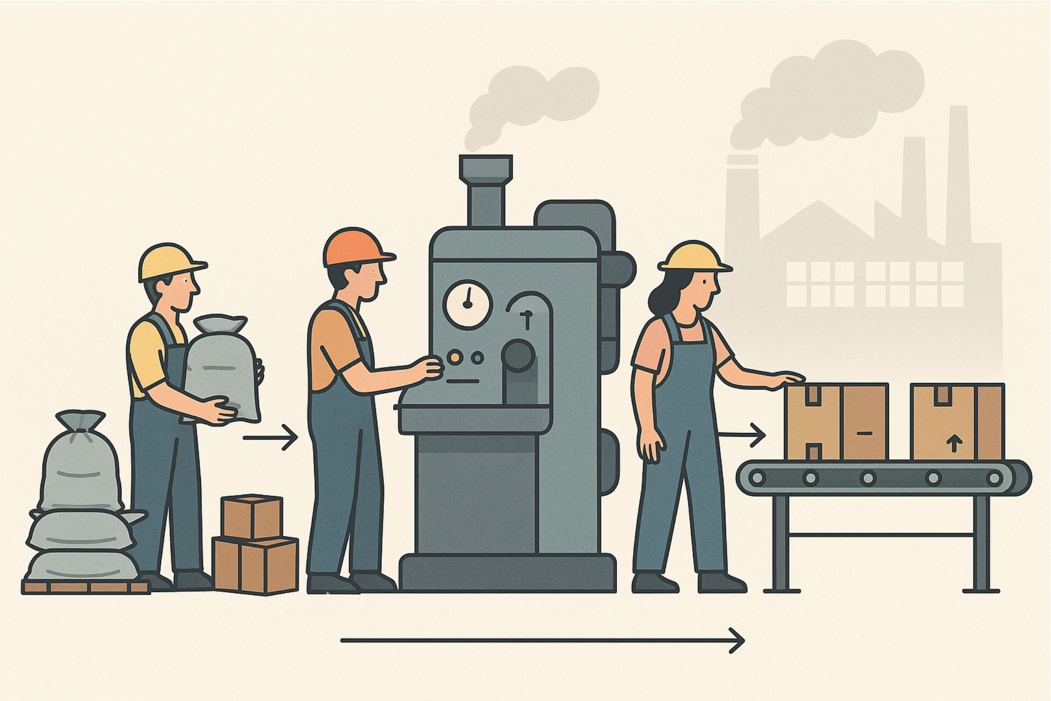


Year: 1981
Global percentile
100
75
50
25
0
0
25
50
75
100
Country percentile
Country
BRA
CHN
FRA
IND
MEX
USA
Based on PovcalNet data from World Bank. By @billywoom.

The continuous technological revolution

The Technological Revolution

Technology: The description of a process using a set of materials and other inputs, including the work of people and machines, to produce an output.

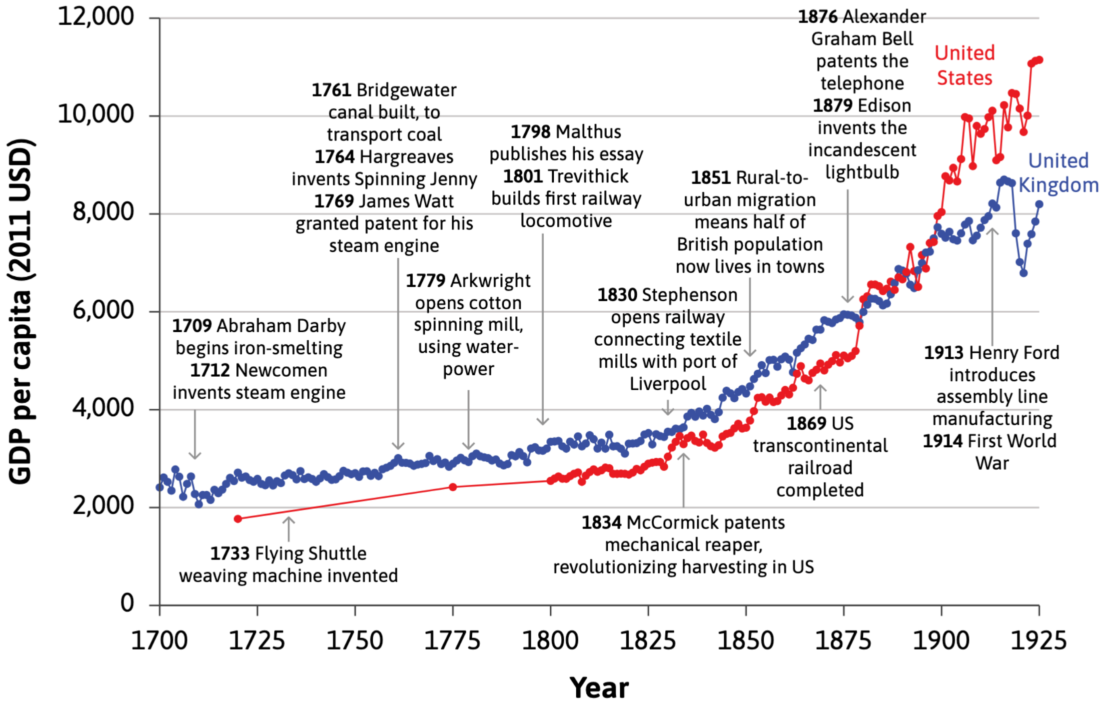


By reducing the amount of work-time it takes to produce the things we need, technological changes allowed significant increases in living standards.

The Industrial Revolution

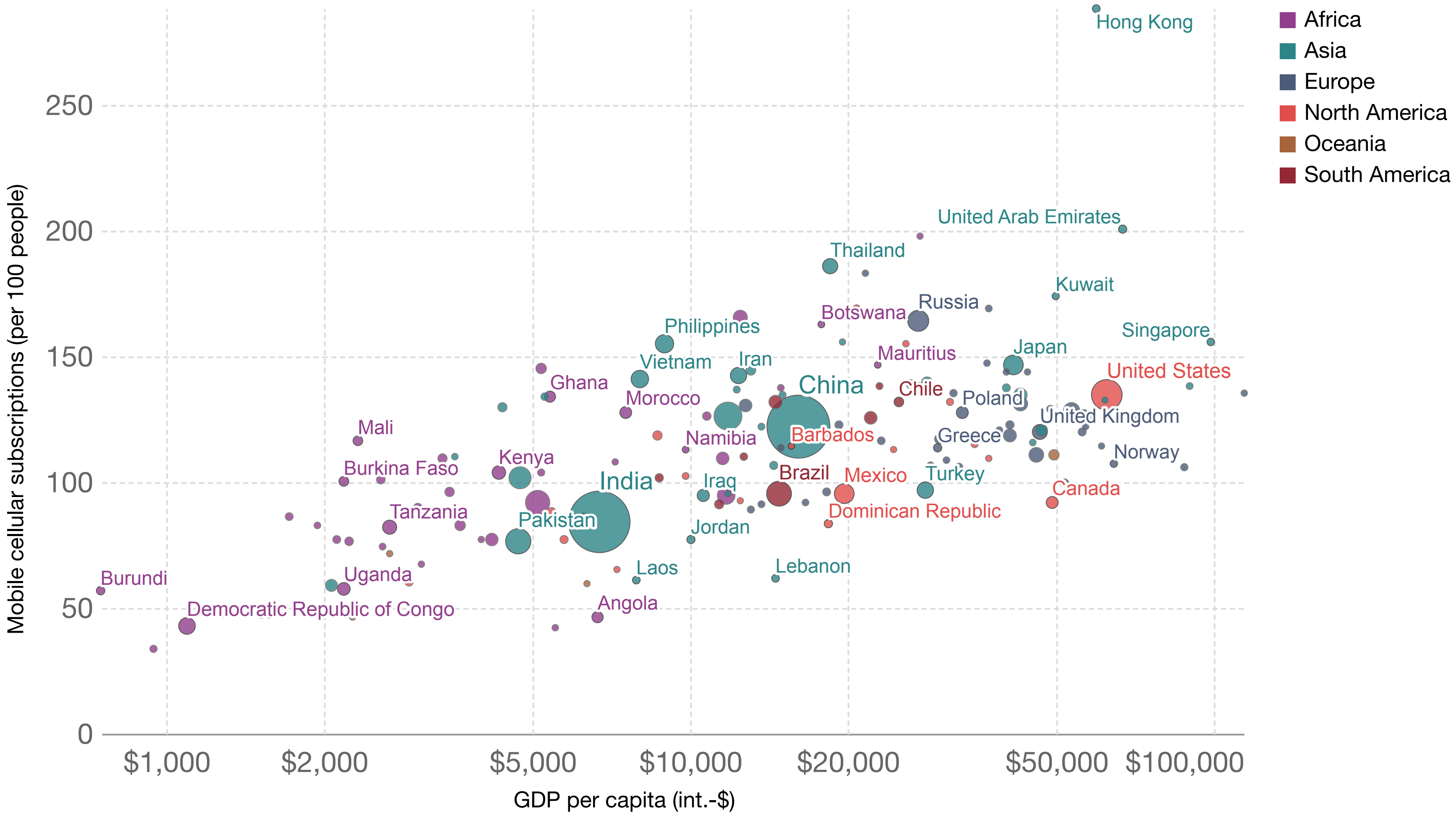
Remarkable scientific and technological advances occurred more or less at the same time as the upward kink in the hockey stick in Britain in the middle of the 18th century.

Industrial Revolution: a wave of technological advances starting in Britain in the 18th century, which transformed an agrarian and craft-based economy into a commercial and industrial economy.



Mobile phone subscriptions vs. GDP per capita, 2019

Number of mobile phone subscriptions, measured per 100 people versus gross domestic product (GDP) per capita, measured in constant international-$.



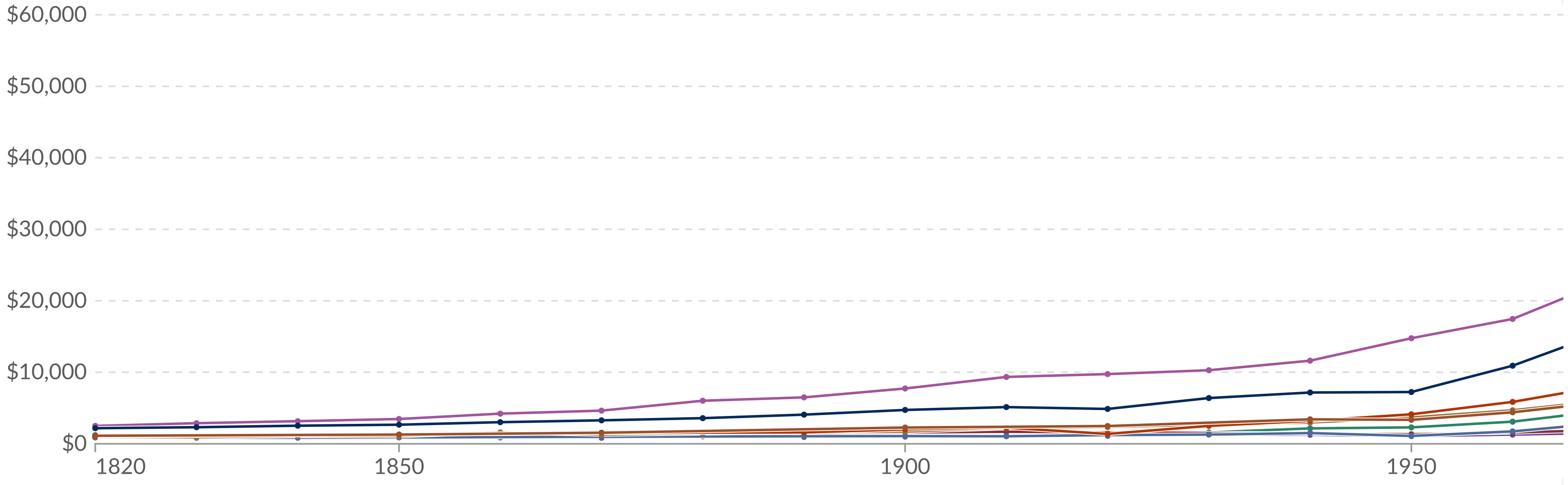
Source: Data compiled from multiple sources by World Bank

Explaining the flat part of the hockey stick

GDP per capita, 1820 to 2022

GDP per capita is a country's gross domestic product divided by its population. This data is adjusted for inflation and differences in living costs between countries.

Table Map Line +2 Edit countries and regions Settings



Play time-lapse 1820 2022

Data source: Bolt and van Zanden – Maddison Project Database 2023 – Learn more about this data

Note: This data is expressed in international-$ at 2011 prices.

Download Share Donate Explore the data →

Diminishing average product of labour

An agricultural economy

Factors of production: labour and land

- production: one good, grain.
- only farm labour, working on the land.

Ceteris paribus assumption: that the amount of land is fixed and all of the same quality.

Average productivity

$$average\ productivity\ of\ labour = \frac{total\ output}{total\ number\ of\ farmers}$$

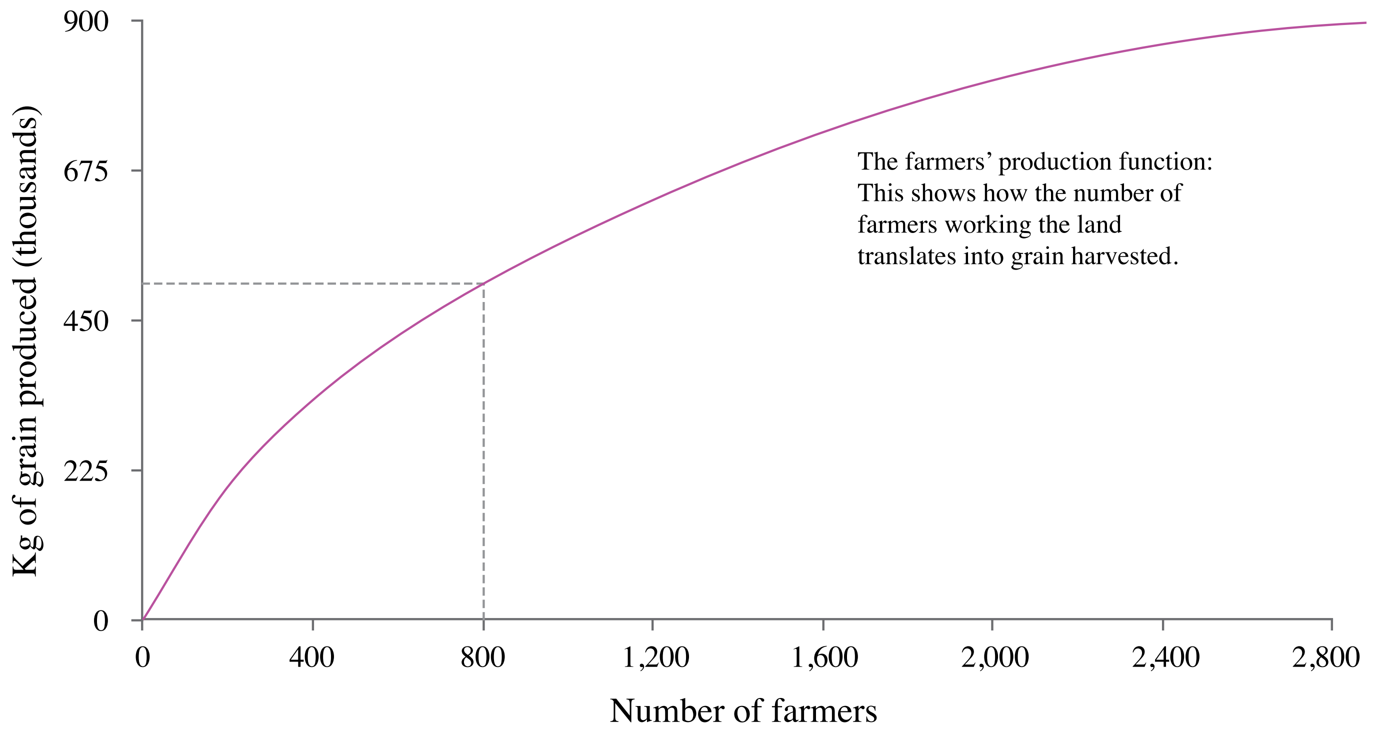
Production function: relationship between the amount of output produced and the amounts of inputs used to produce it.

$$Y = f(X)$$

Production function gives maximum output for a given set of inputs.

Diminishing average product of labour

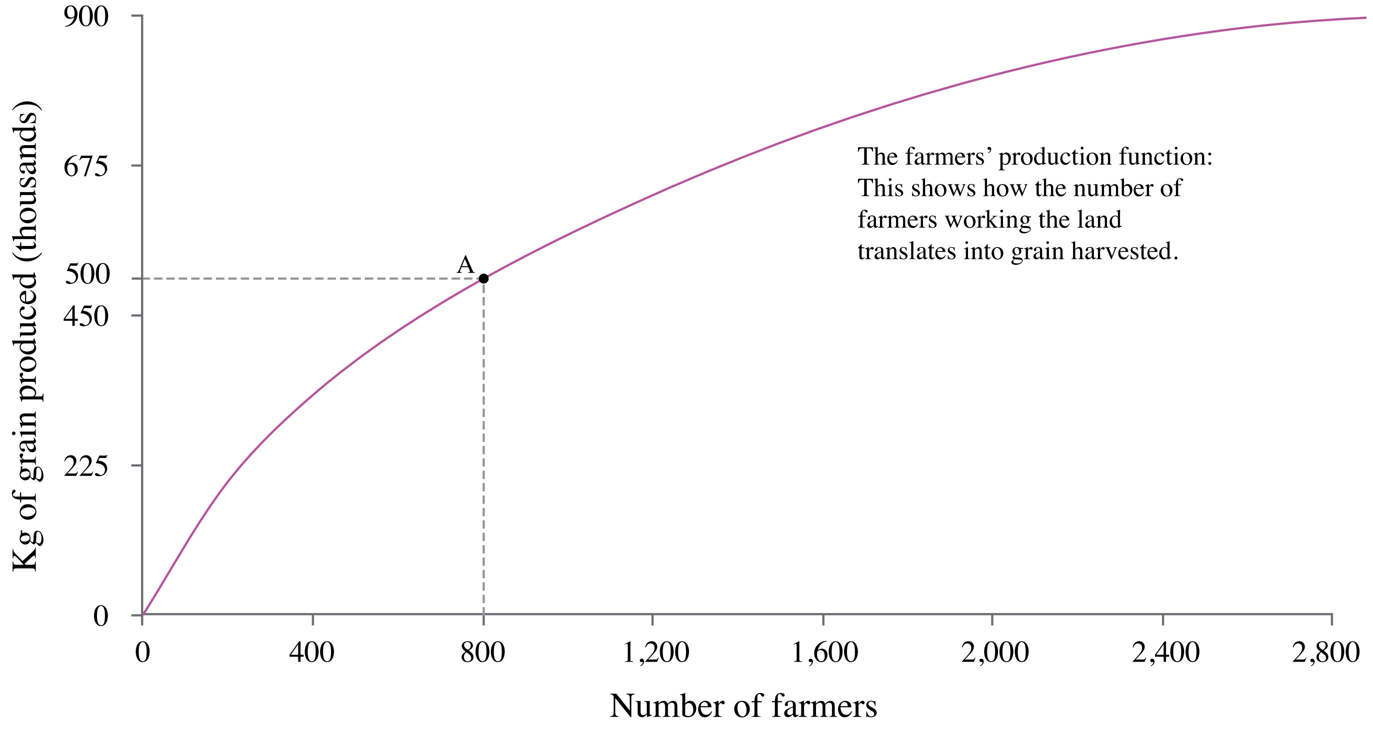
If we hold one input (land) fixed, and expand the other input (labour), the average output per worker is going to fall.



The production function shows how the number of farmers working the land translates into grain produced at the end of the growing season.

Diminishing average product of labour

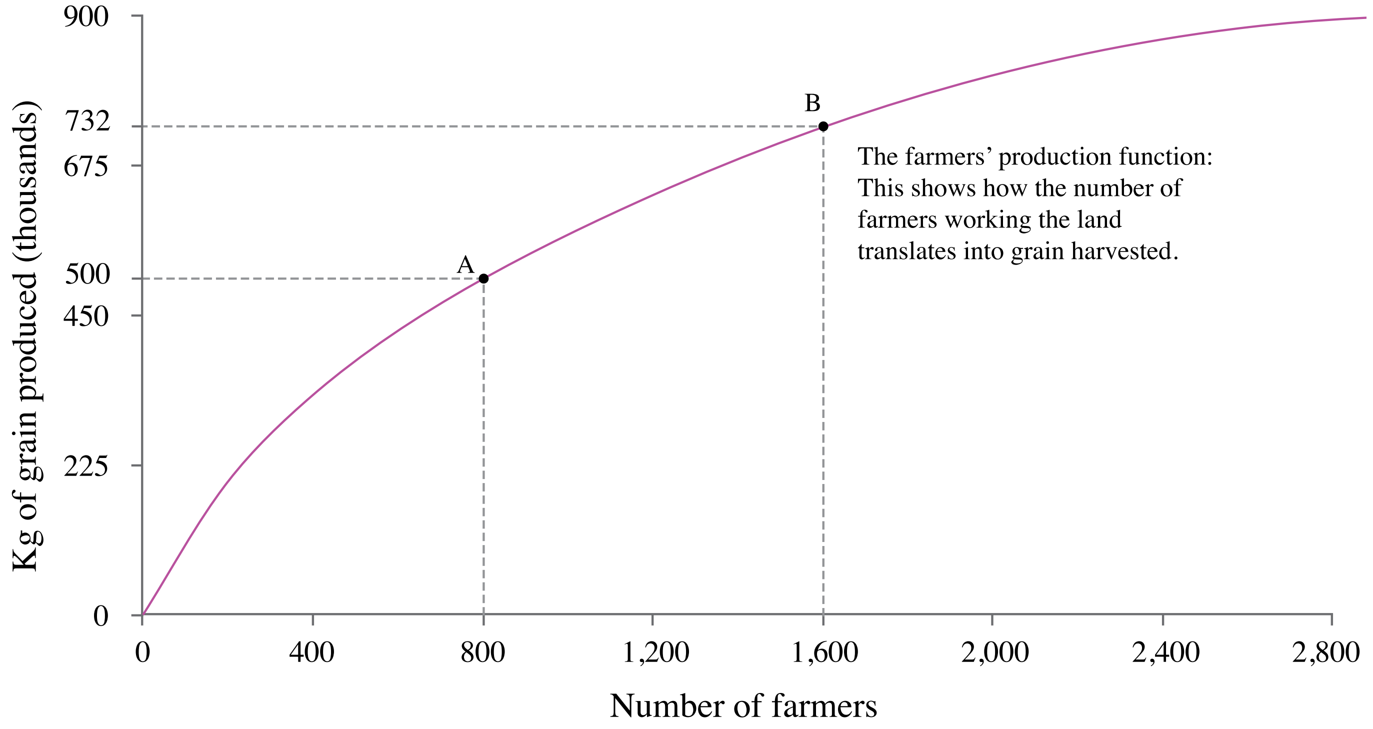
If we hold one input (land) fixed, and expand the other input (labour), the average output per worker is going to fall.



Point A on the production function shows the output of grain produced by 800 farmers.

Diminishing average product of labour

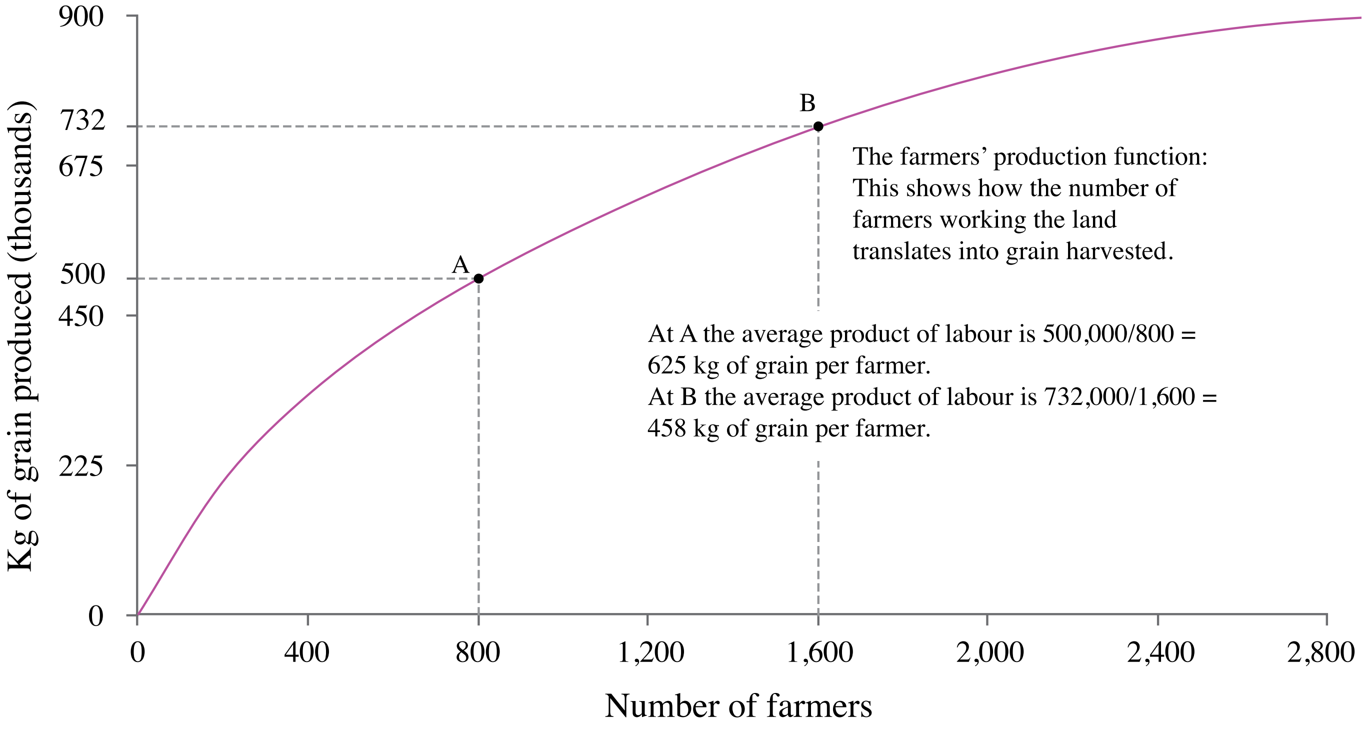
If we hold one input (land) fixed, and expand the other input (labour), the average output per worker is going to fall.



Point B on the production function shows the amount of grain produced by 1,600 farmers.

Diminishing average product of labour

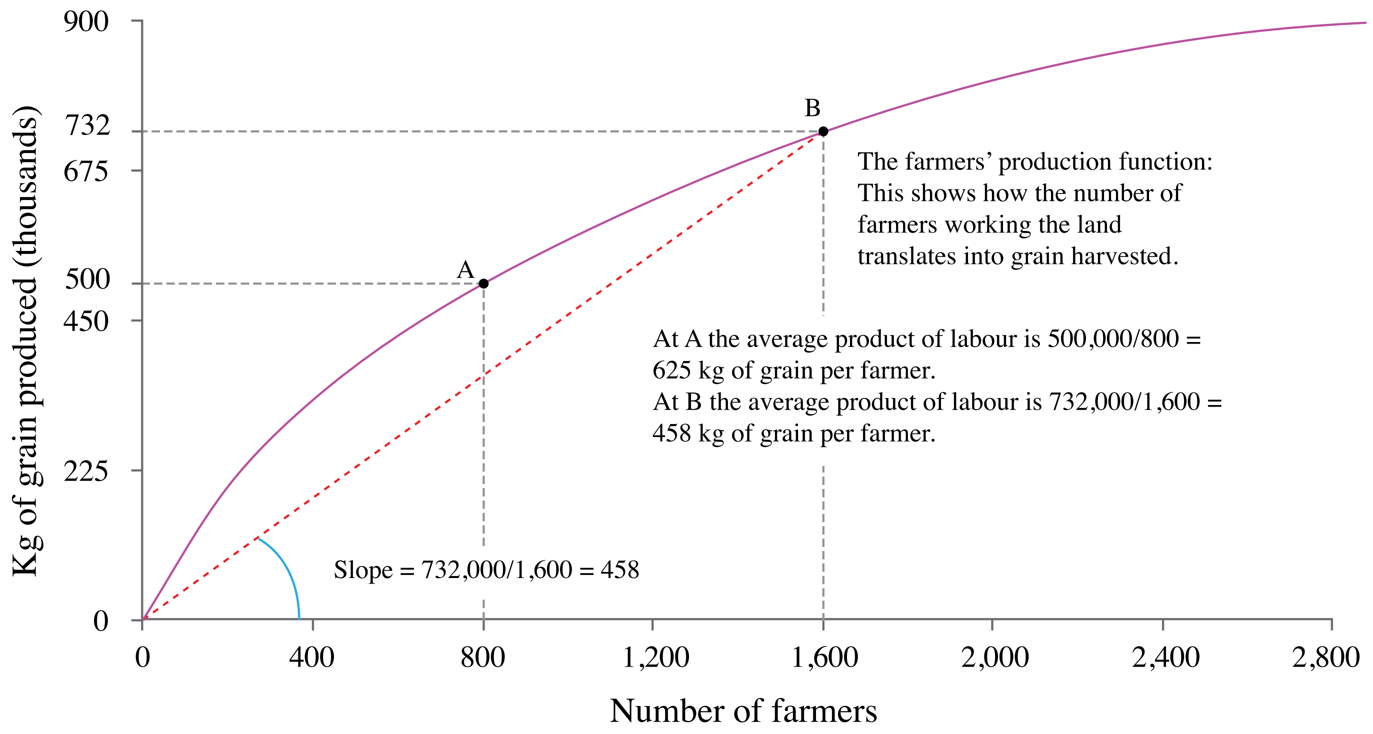
If we hold one input (land) fixed, and expand the other input (labour), the average output per worker is going to fall.



At A, the average product of labour is 500,00/800 = 625 kg of grain per farmer. At B, the average product of labour is 732,000/1,600 = 458 kg of grain per farmer.

Diminishing average product of labour

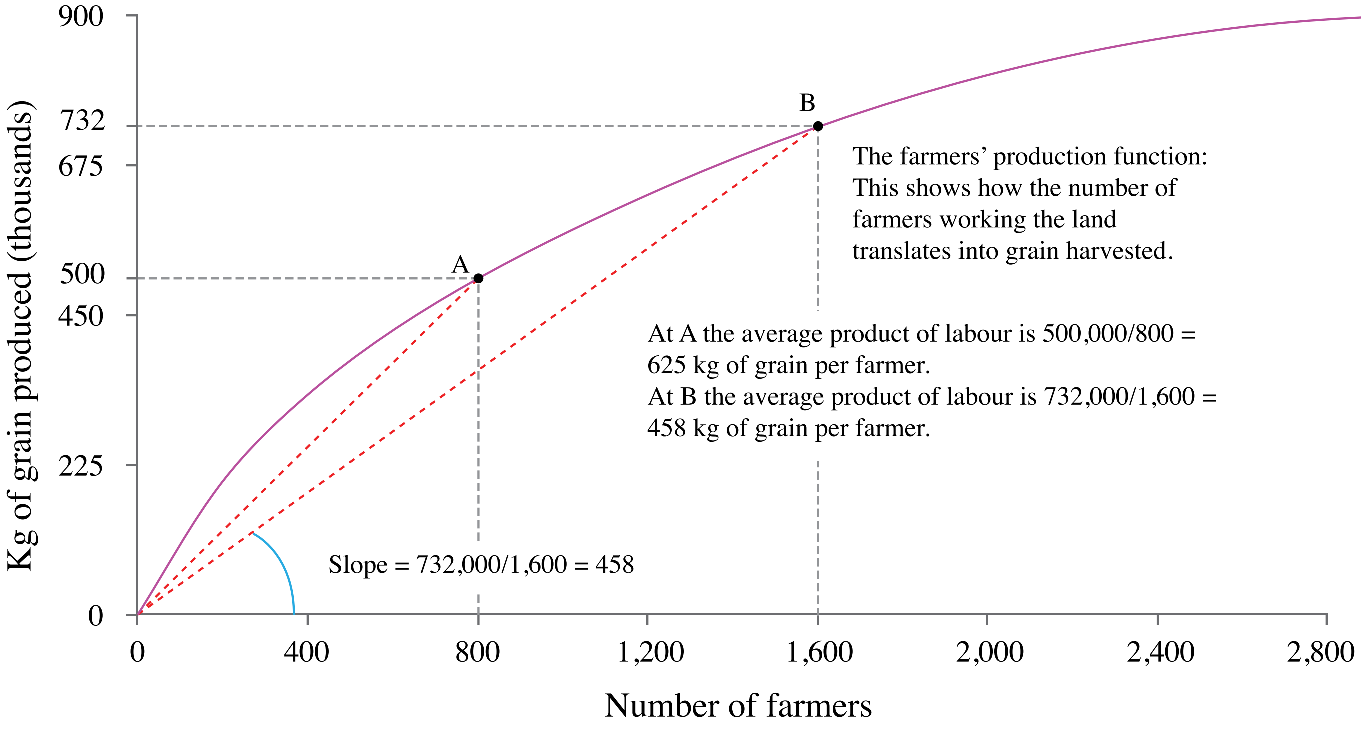
If we hold one input (land) fixed, and expand the other input (labour), the average output per worker is going to fall.



The slope of the ray from the origin to point B on the production function shows the average product of labour at point B. The slope is 458, meaning an average product of 458 kg per farmer when 1,600 farmers work the land.

Diminishing average product of labour

If we hold one input (land) fixed, and expand the other input (labour), the average output per worker is going to fall.



The slope of the ray to point A is steeper than to point B. When only 800 farmers work the land there is a higher average product of labour. The slope is 625, the average product of 625 kg per farmer that we calculated previously.

The Malthusian trap, population, and the average product of labour

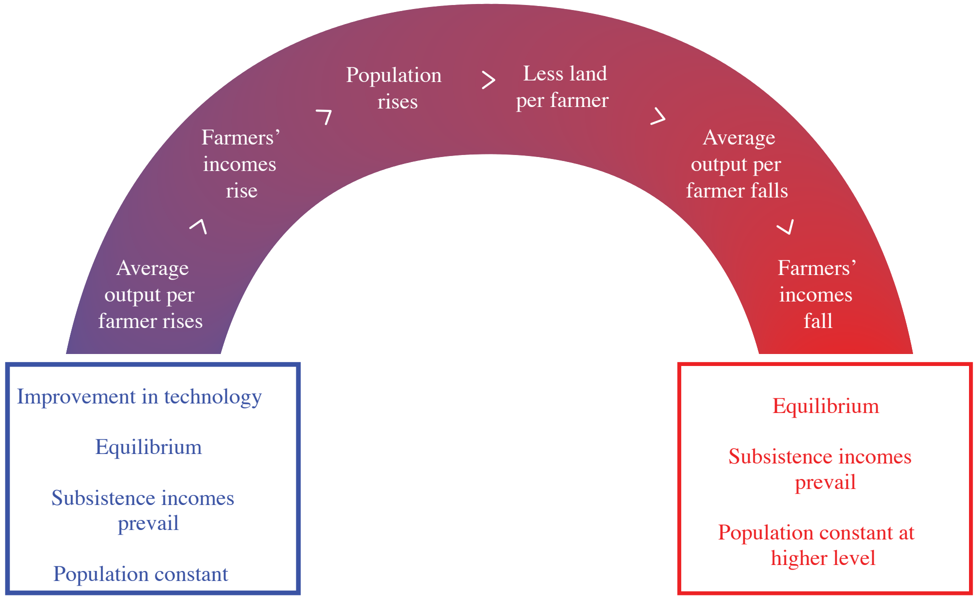
Malthus argument

1. Population expands if living standards increase
2. Law of diminishing average product of labour implies that as more people work on the land, their income will inevitably fall

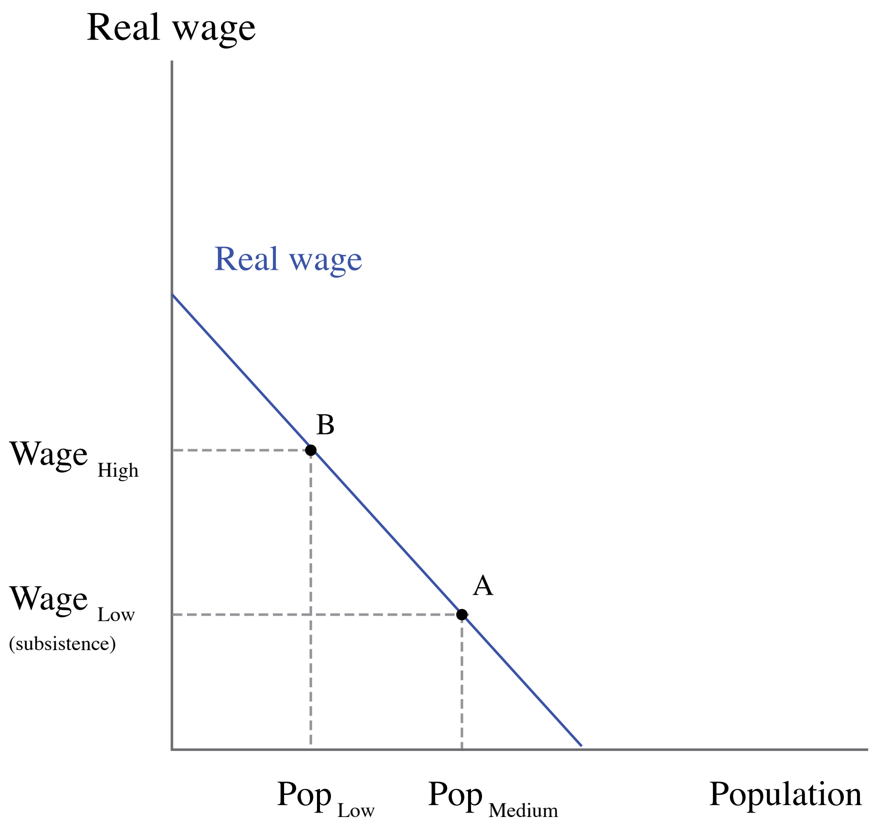
In equilibrium, living standards will be forced down to subsistence level.

Population and income will stay constant.

Self-correcting response to new technology. In the long run, an increase in productivity will result in increased population but not increased wages.

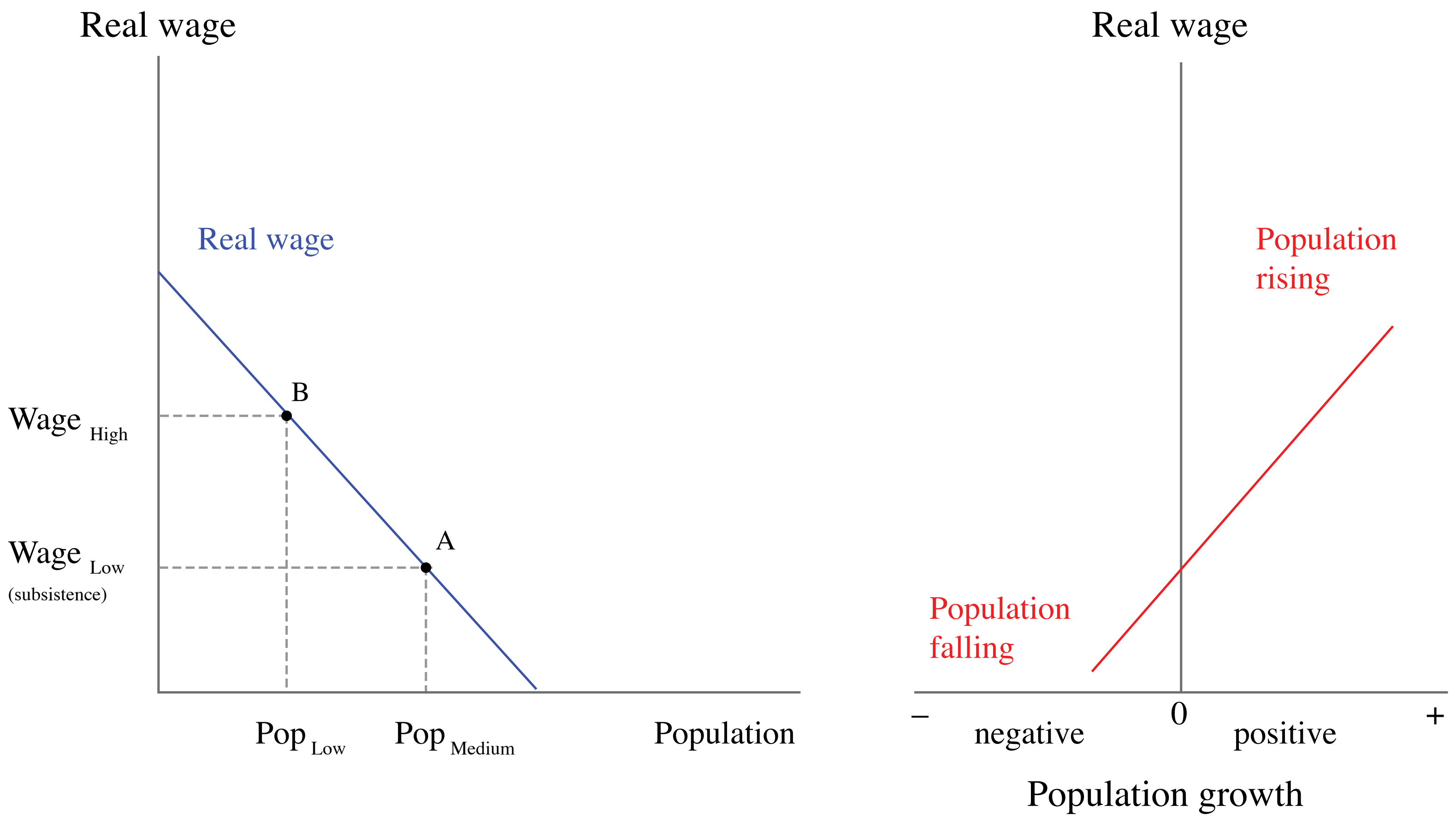


Modelling Malthus



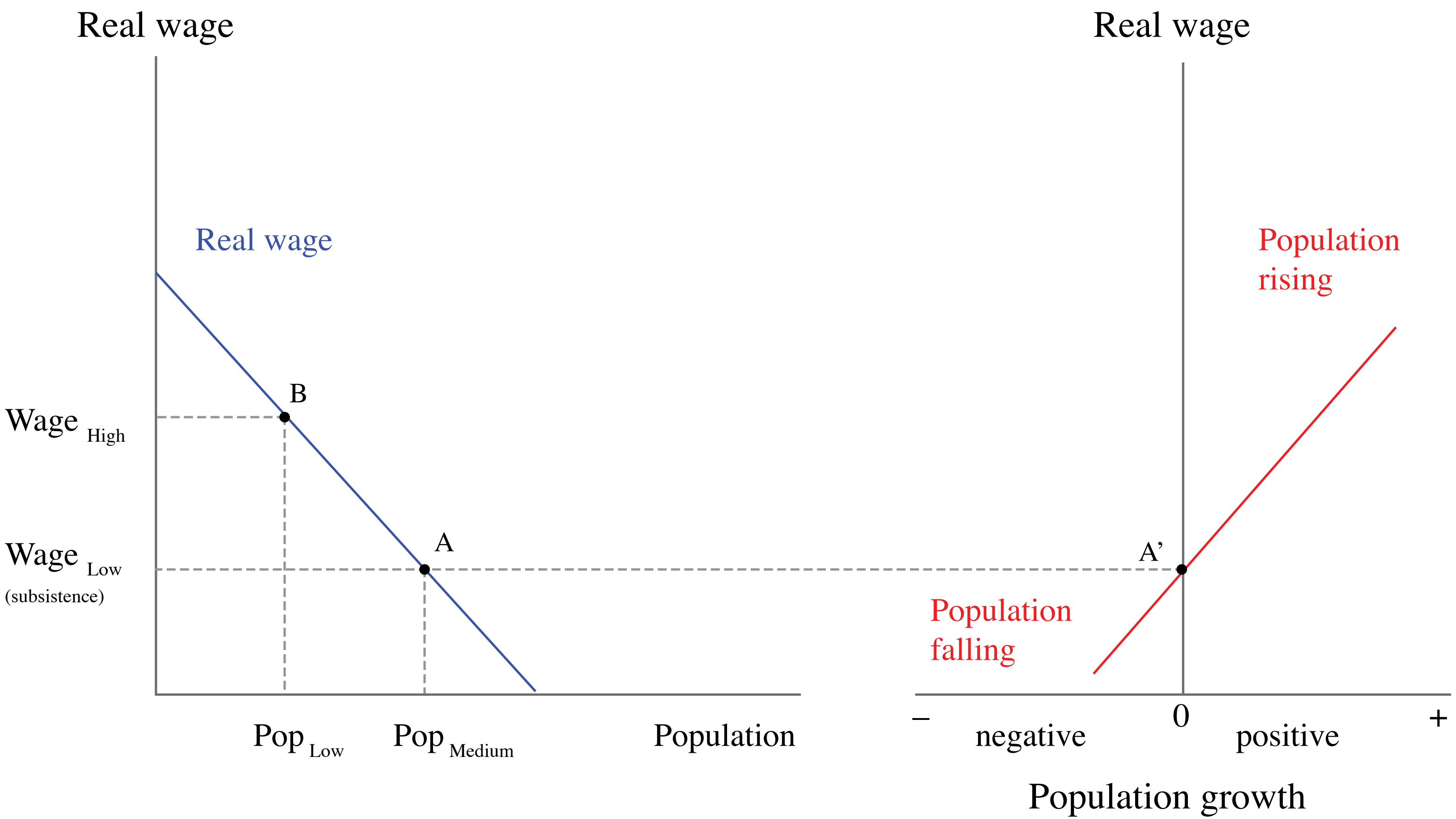
At a medium population level, the wage of people who work the land is at subsistence level (point A). The wage is higher at point B, where the population is smaller, because the average product of labour is higher.

Modelling Malthus



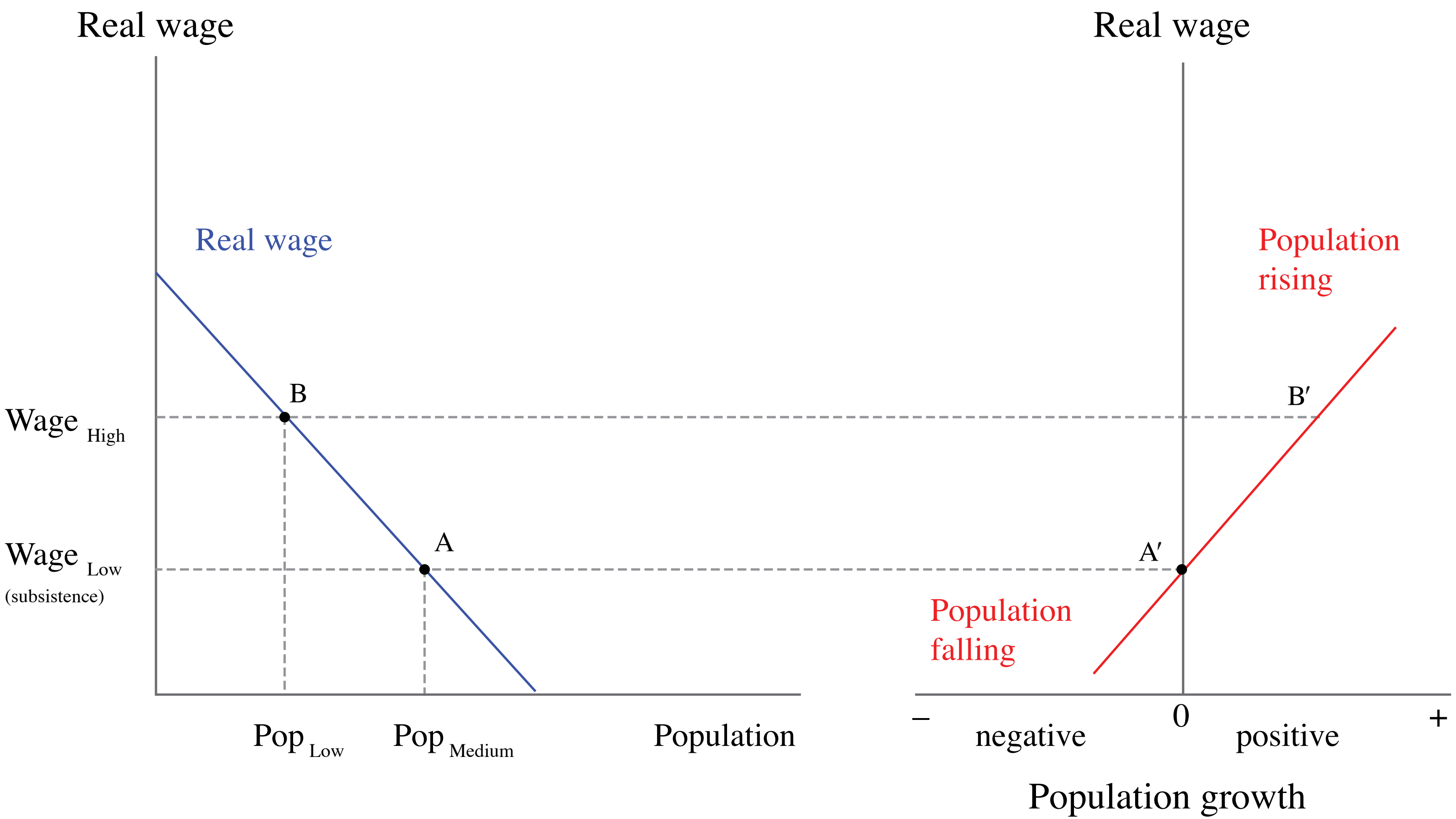
When wages (on the vertical axis) are high, population growth (on the horizontal axis) is positive (so the population will rise). When wages are low, population growth is negative (population falls).

Modelling Malthus



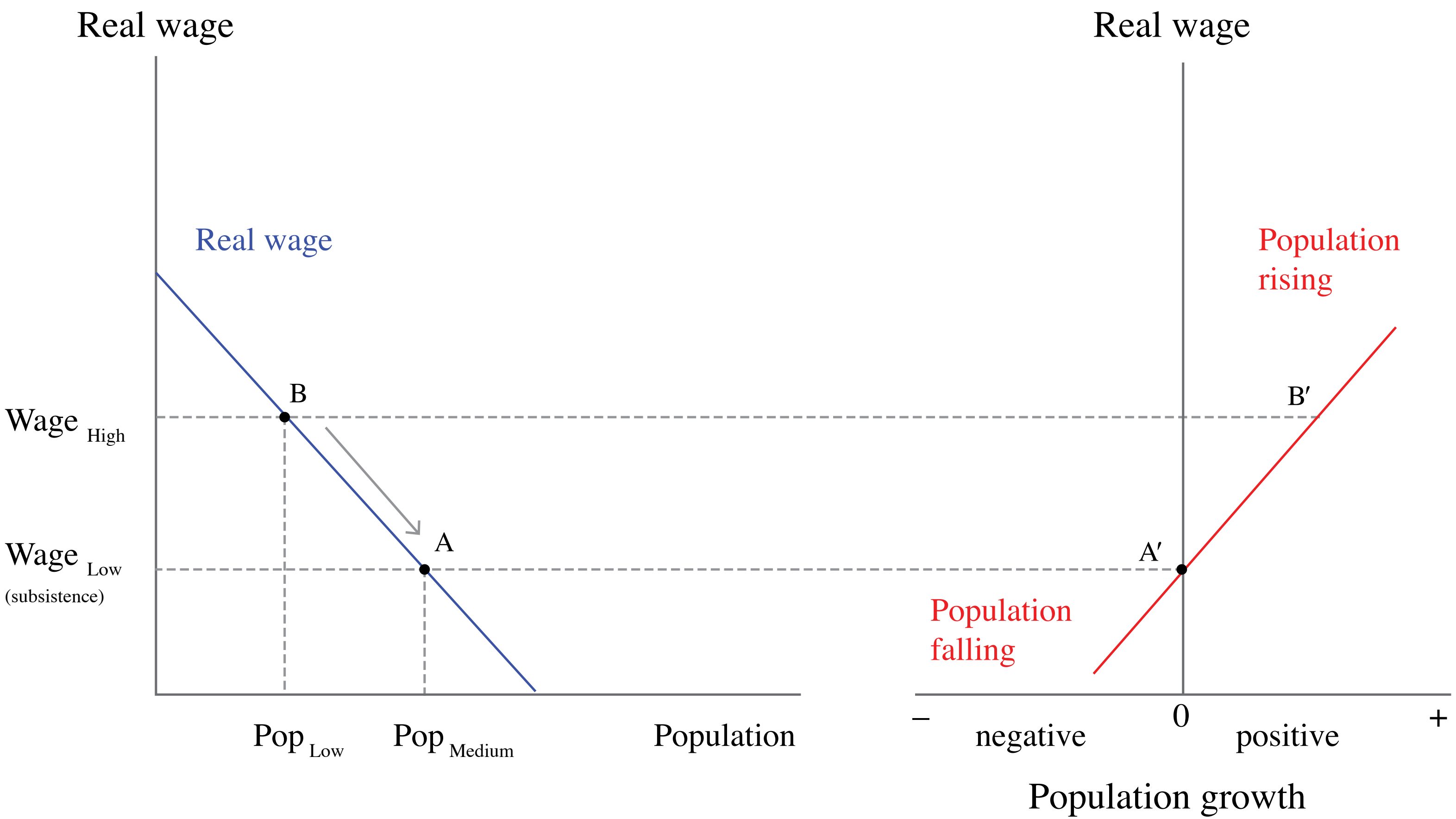
At point A, on the left, population is medium-sized and the wage is at subsistence level. So if the economy is at point A, it is in equilibrium: population stays constant and wages remain at subsistence level.

Modelling Malthus



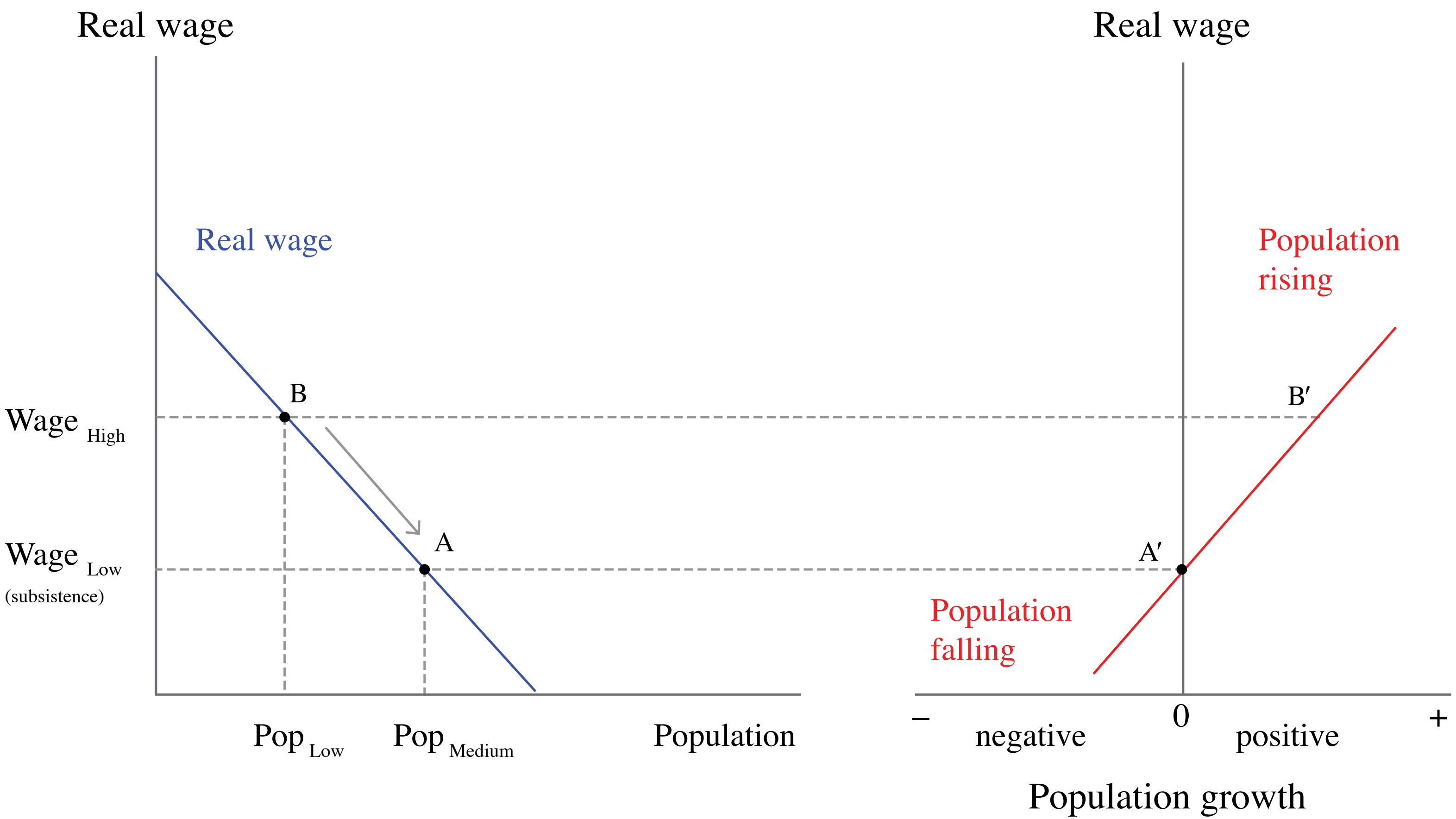
Suppose the economy is at B, with a higher wage and lower population. Point B′, on the right, shows that the population will be rising.

Modelling Malthus



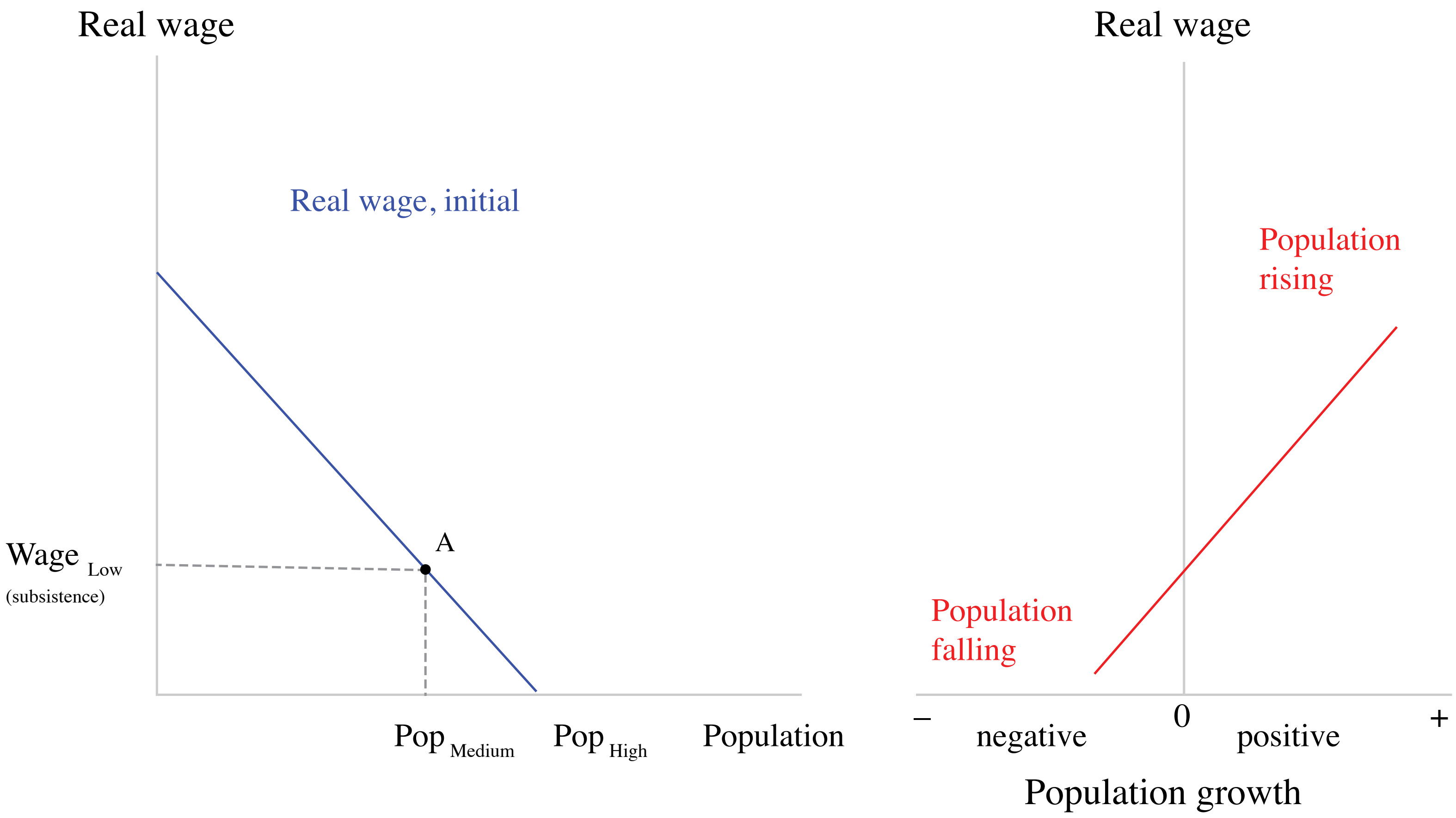
As the population rises, the economy moves down the line in the left diagram: wages fall until they reach equilibrium at A.

Modelling Malthus



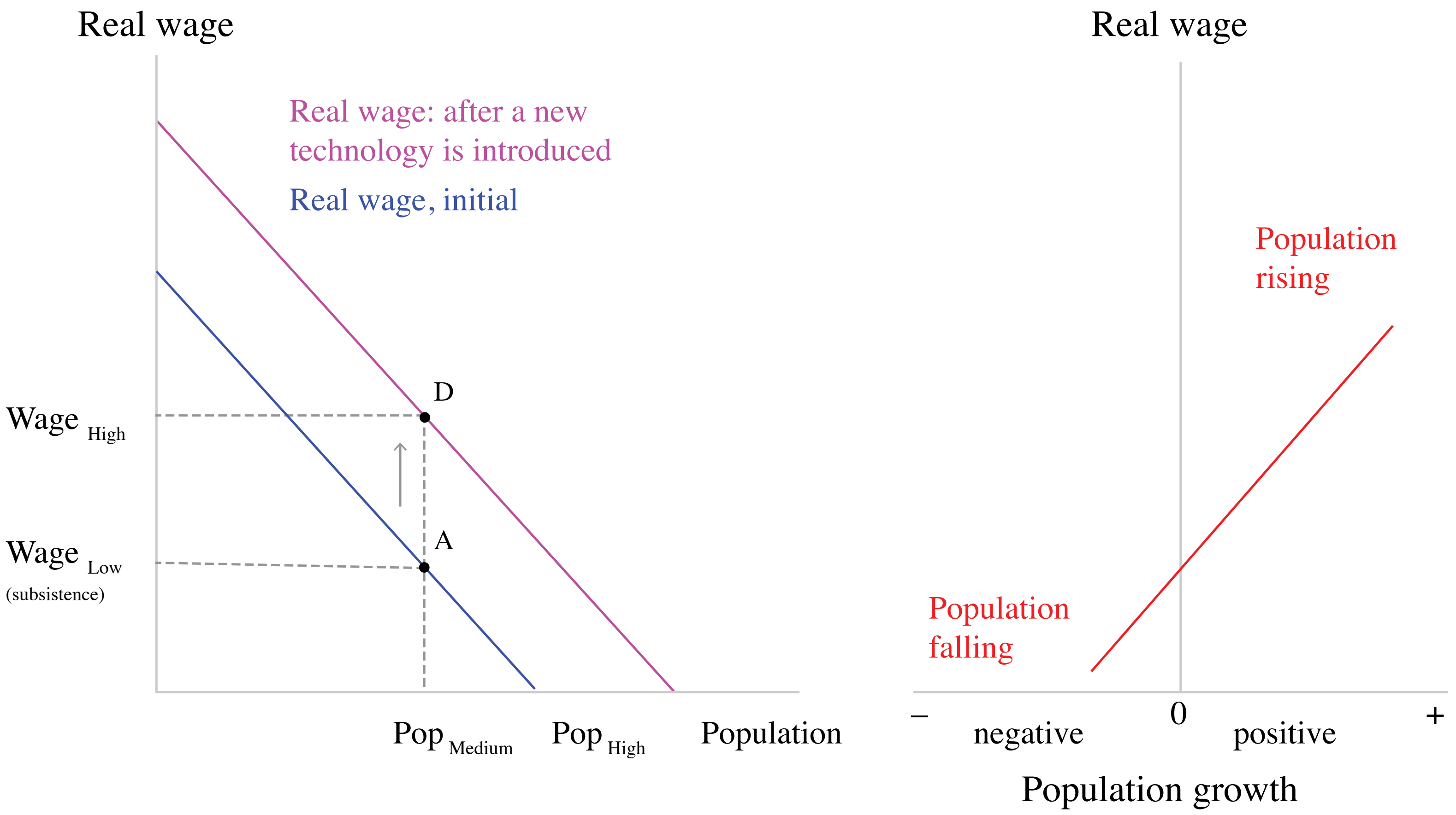
Malthusian population trap: Population will be constant when the wage is at subsistence level, it will rise when the wage is above subsistence level, and it will fall when the wage is below subsistence level.

Modelling Malthus: productivity increases



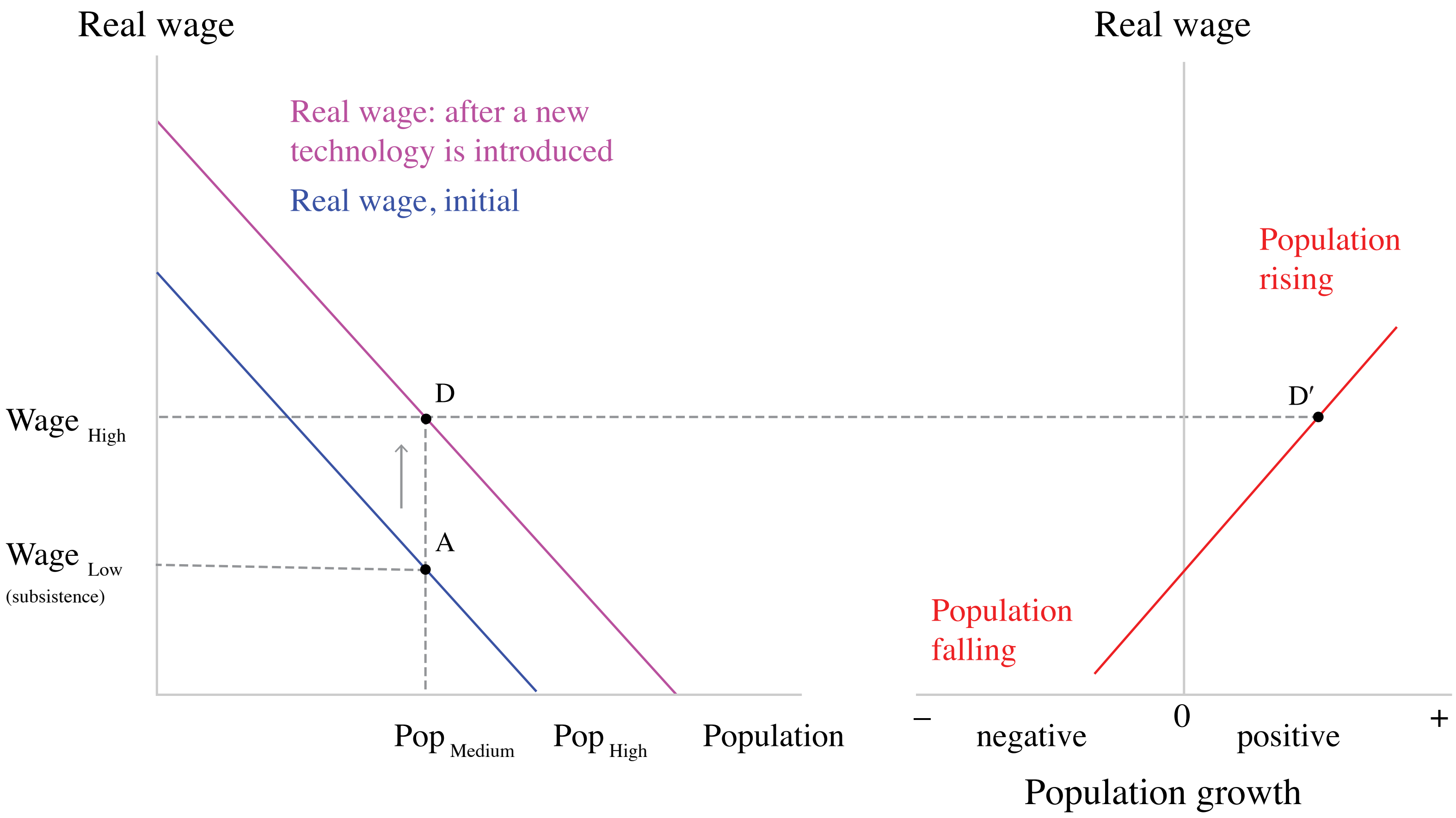
The economy starts at point A, with a medium-sized population and wage at subsistence level.

Modelling Malthus: productivity increases



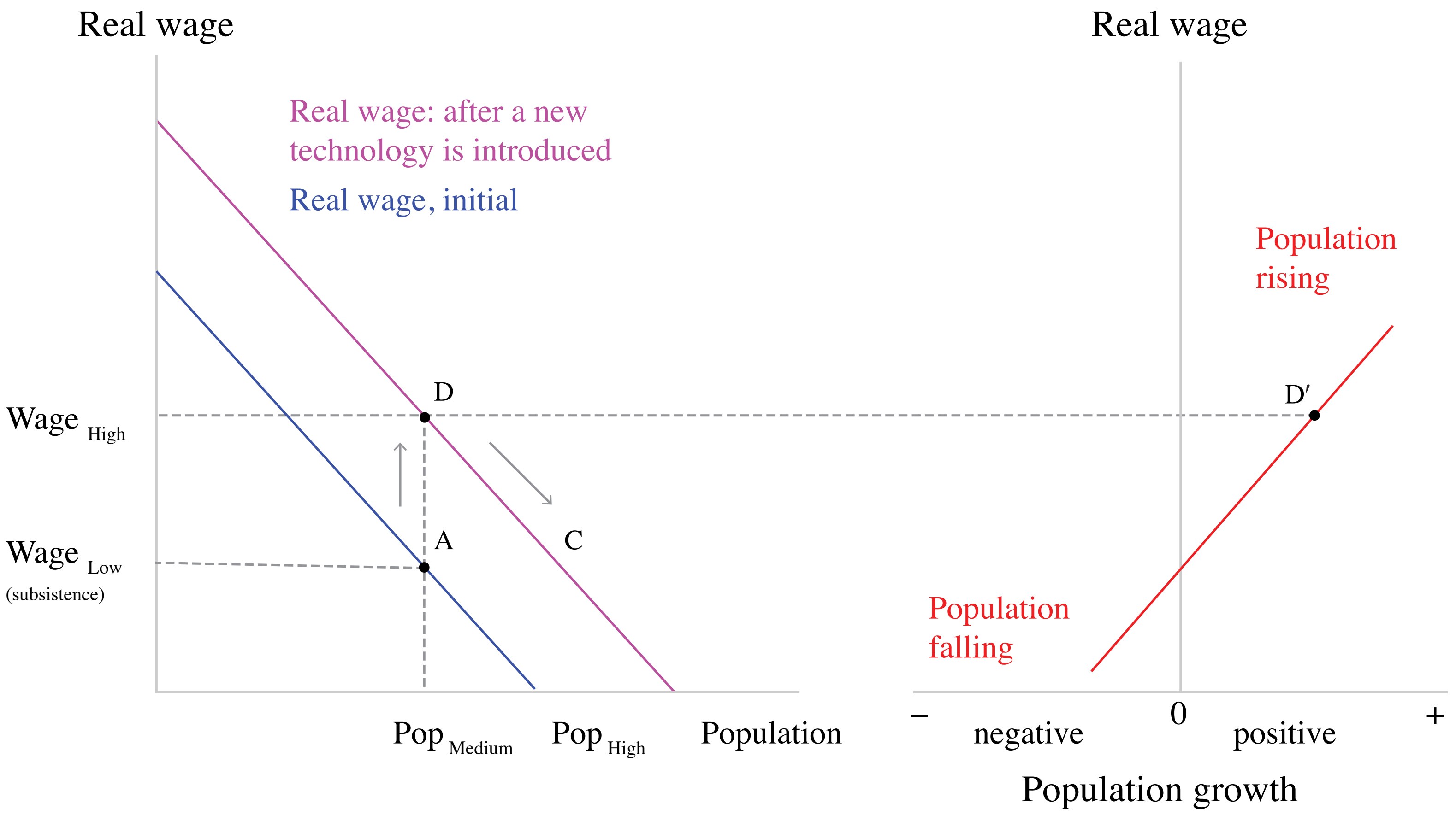
Technological improvement (better seeds) raises the average product of labour, and the wage is higher for any level of population. The real wage line shifts upward. At the initial population level, the wage increases and the economy moves to point D.

Modelling Malthus: productivity increases



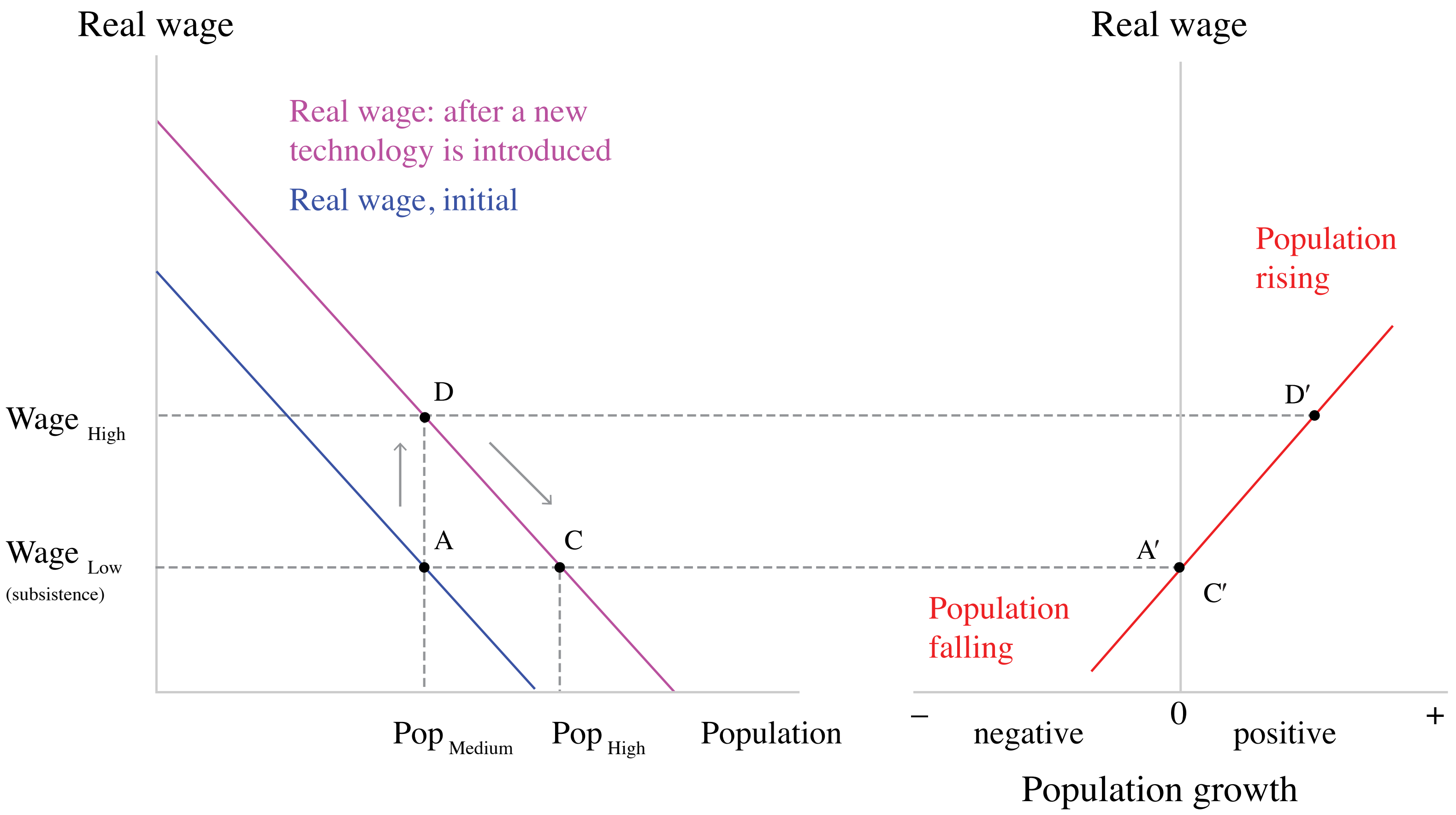
At point D, the wage has risen above subsistence level and therefore the population starts to grow (point D′).

Modelling Malthus: productivity increases



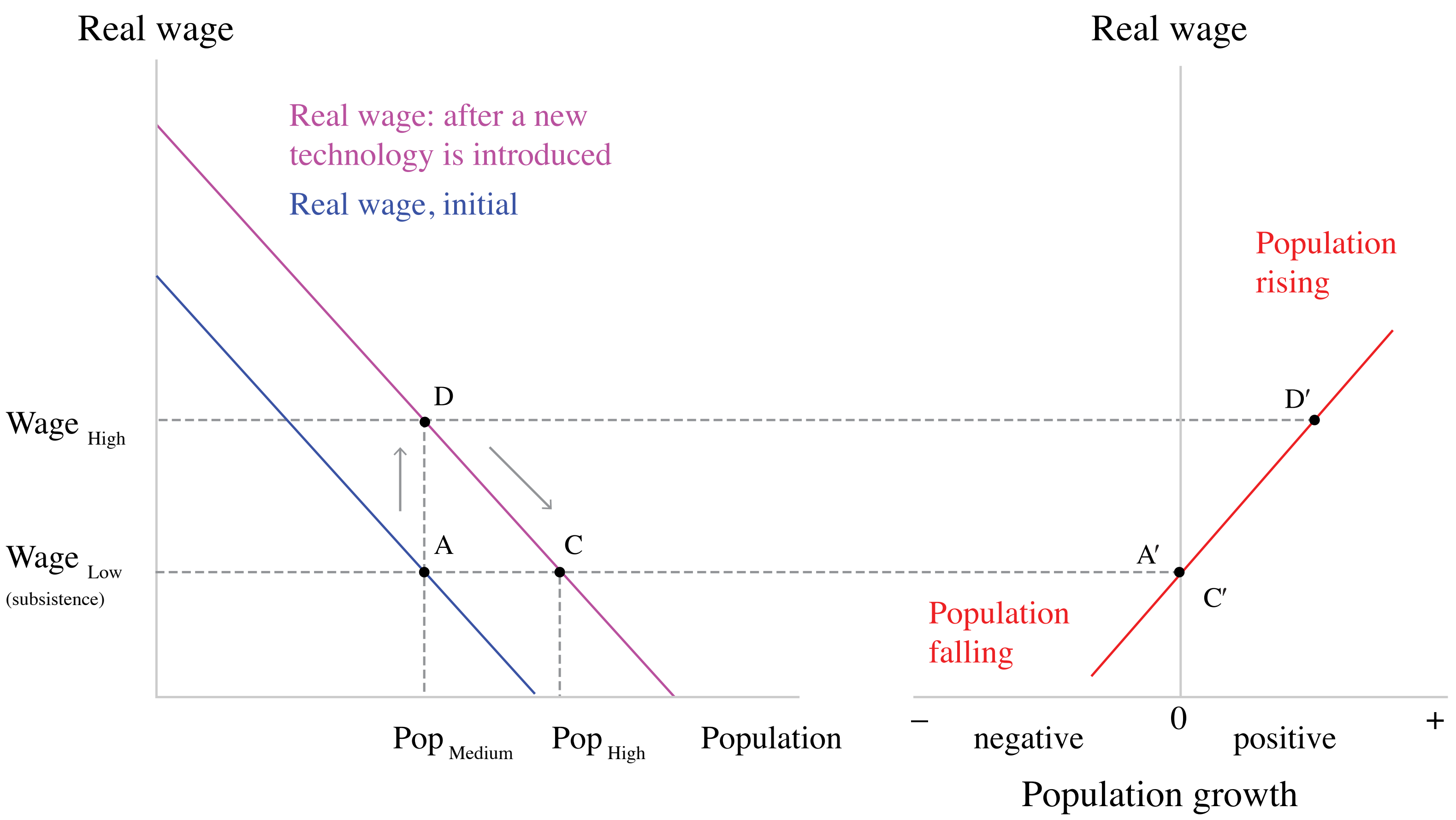
As population rises, the wage falls, due to the diminishing average product of labour. The economy moves down the real-wage curve from D.

Modelling Malthus: productivity increases



At C, the wage has reached subsistence level again. The population remains constant (point C′). The population is higher at equilibrium C than it was at equilibrium A.

Modelling Malthus: productivity increases



Malthusian model predicts that even if productivity increases, living standards in the long run do not.

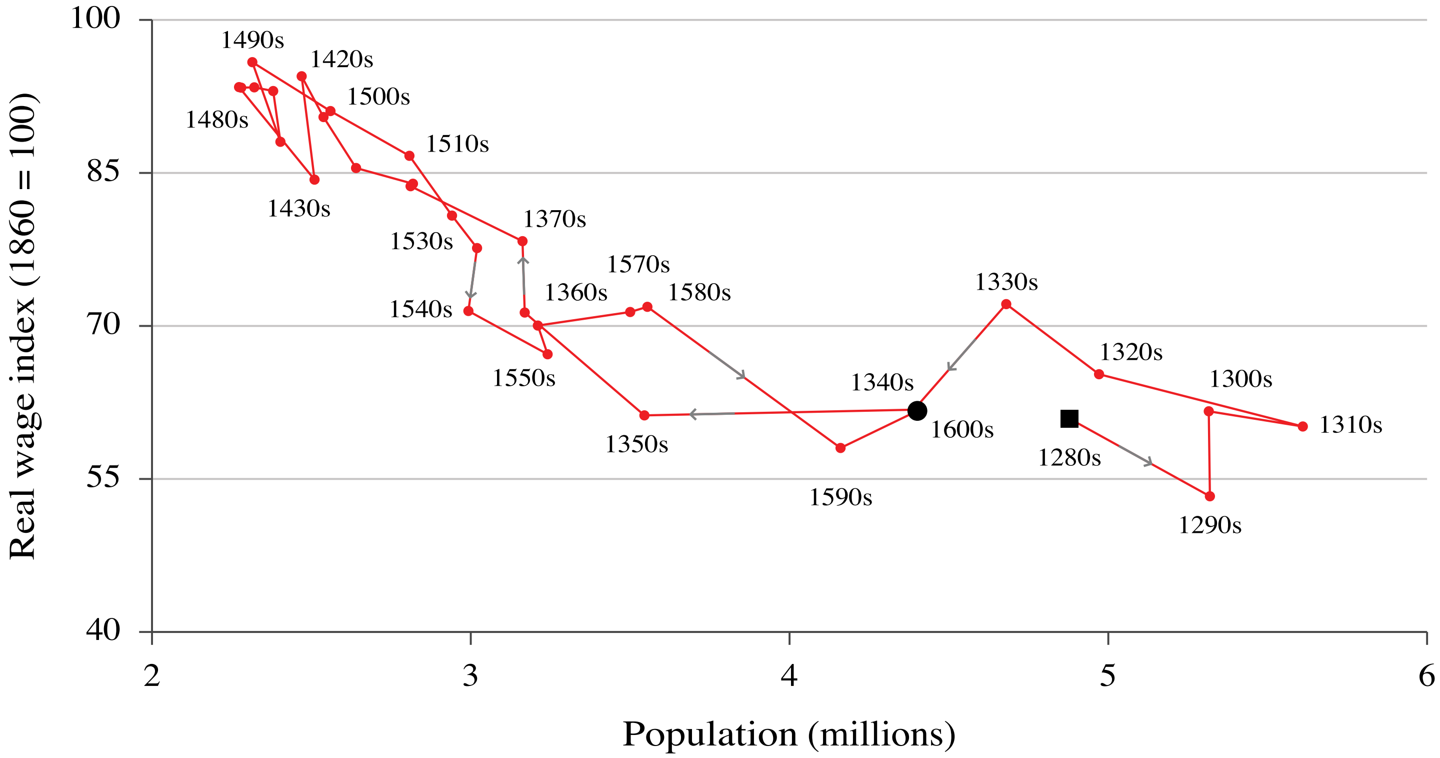
The Malthusian trap and long-term economic stagnation

Testing Malthus

1) Diminishing average product of labour

2) Rising population in response to increases in wages

3) **An absence of improvements in technology to offset the diminishing average product of labour**

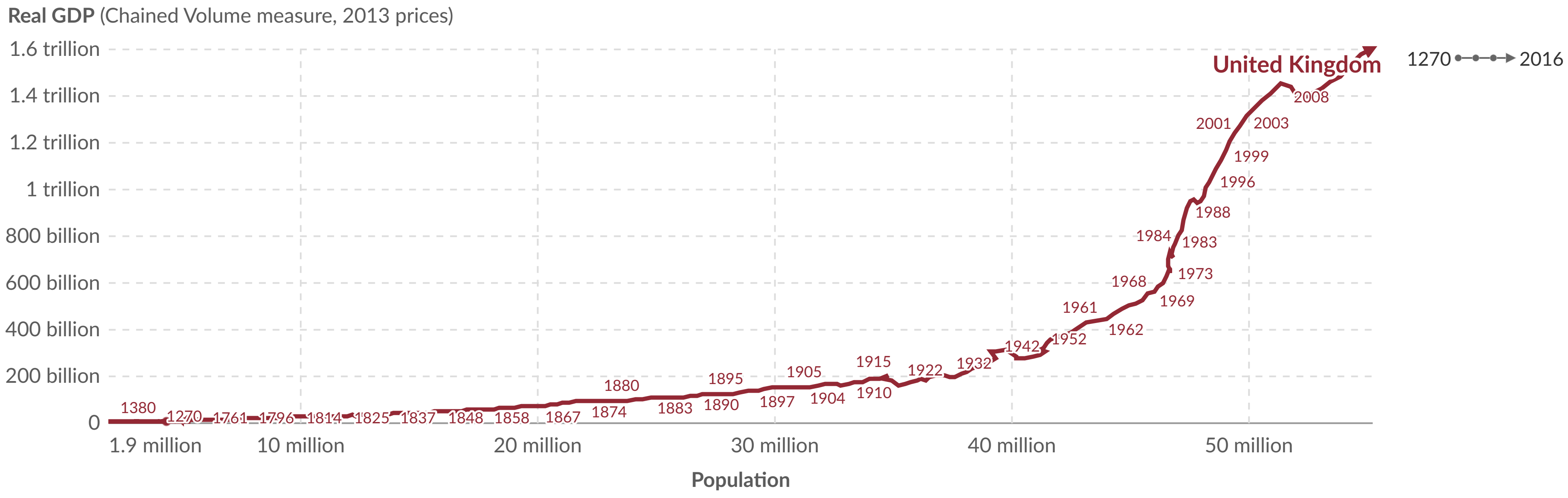


Escaping Malthus

England's economy over the long run: GDP vs. population, United Kingdom, 1270 to 2016

Our World in Data

Table Chart Settings



Play time-lapse 1270 2016

Data source: Broadberry, Campbell, Klein, Overton and van Leeuwen (2015) and Bank of England – Learn more about this data

Explore the data →

Capitalist institutions

Capitalist institutions

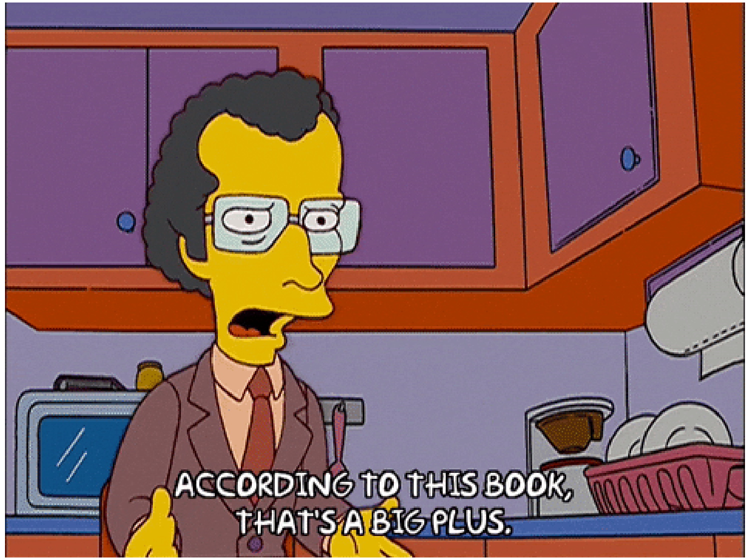
How can we explain the escape from the Malthusian trap?

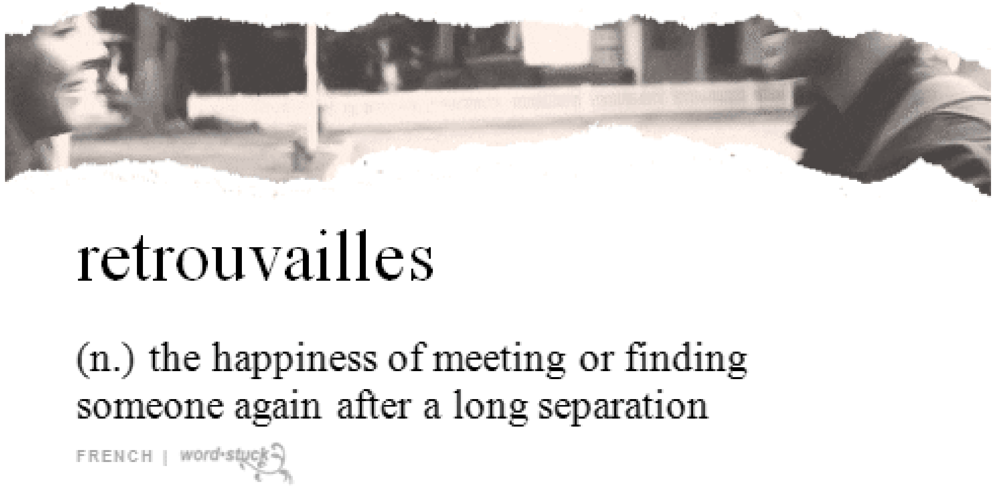
Capitalist revolution: the emergence in the XVIIIth century and eventual global spread of a way of organizing the economy that we now call capitalism

Capitalism: an **economic system** characterized by a particular combination of **institutions**

Institutions

Institutions: the laws and informal rules that regulate social interactions among people and between people and the biosphere, sometimes also termed the rules of the game.





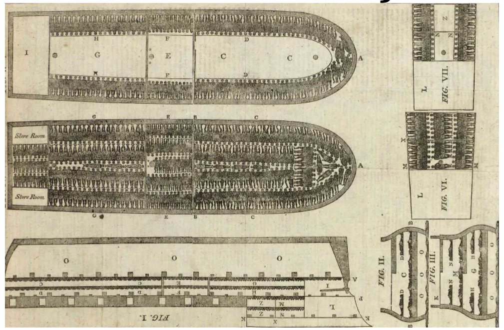
Economic systems

Economic system: a way of organizing the economy that is distinctive in its basic institutions.

Feudalism

Central economic planning

Slave economy

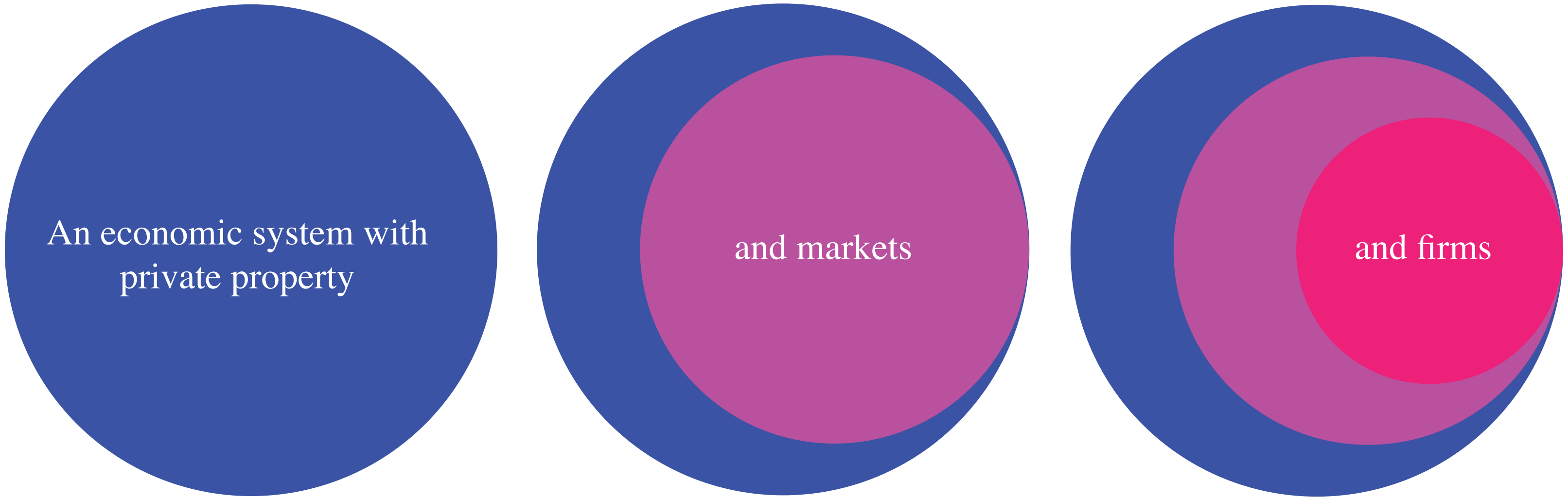


Capitalism

Capitalism

Capitalism: An economic system in which the main form of economic organization is the **firm**, in which the private owners of **capital goods** hire **labour** to produce goods and services for sale on **markets** with the intent of making a profit.

The main economic institutions in a capitalist economic system, then, are **private property**, **markets**, and **firms**.



Self-sufficient family-based production

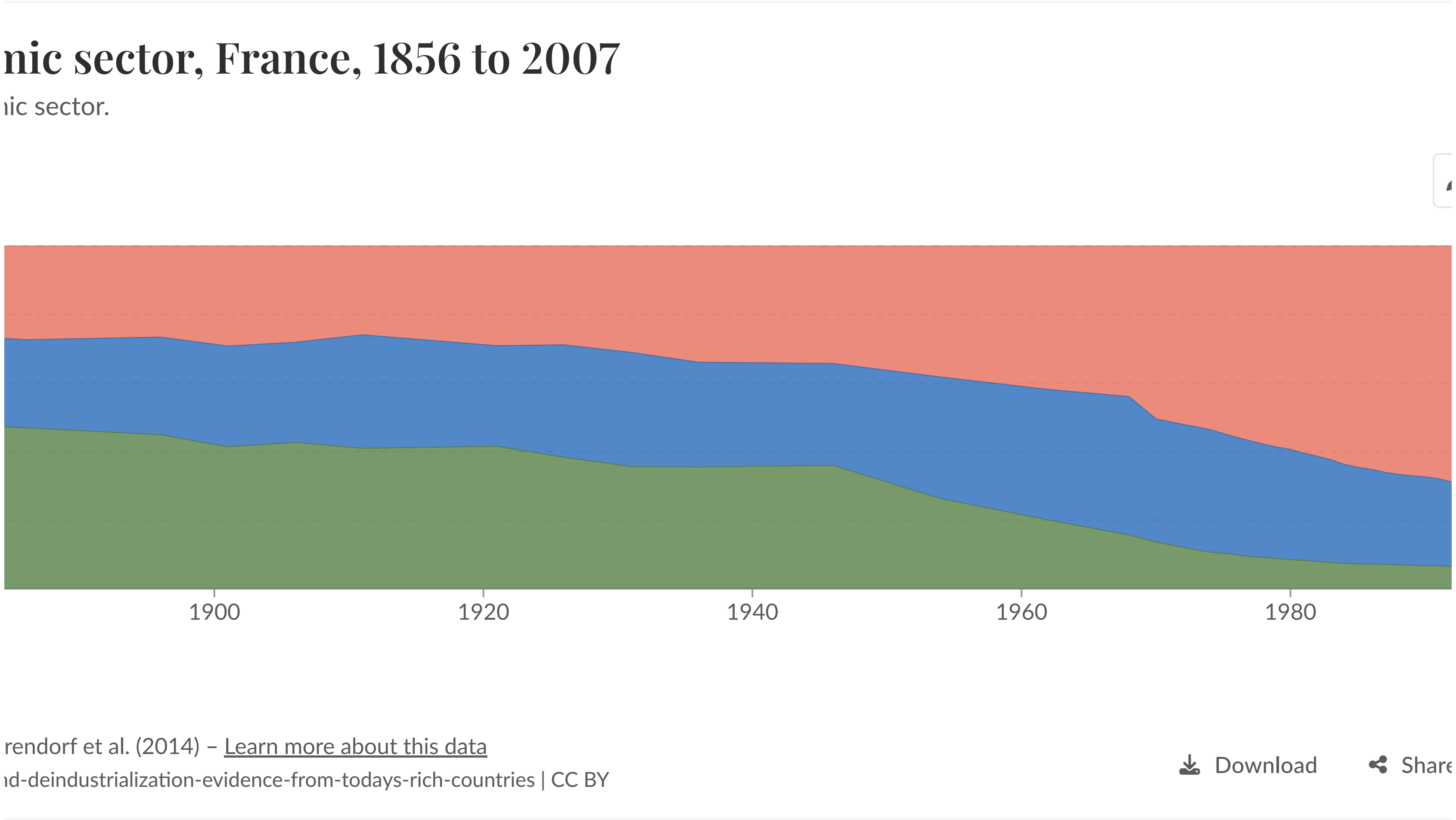
Market economy with family-based production

Capitalist economic system

Structural transformation: From farm to firm

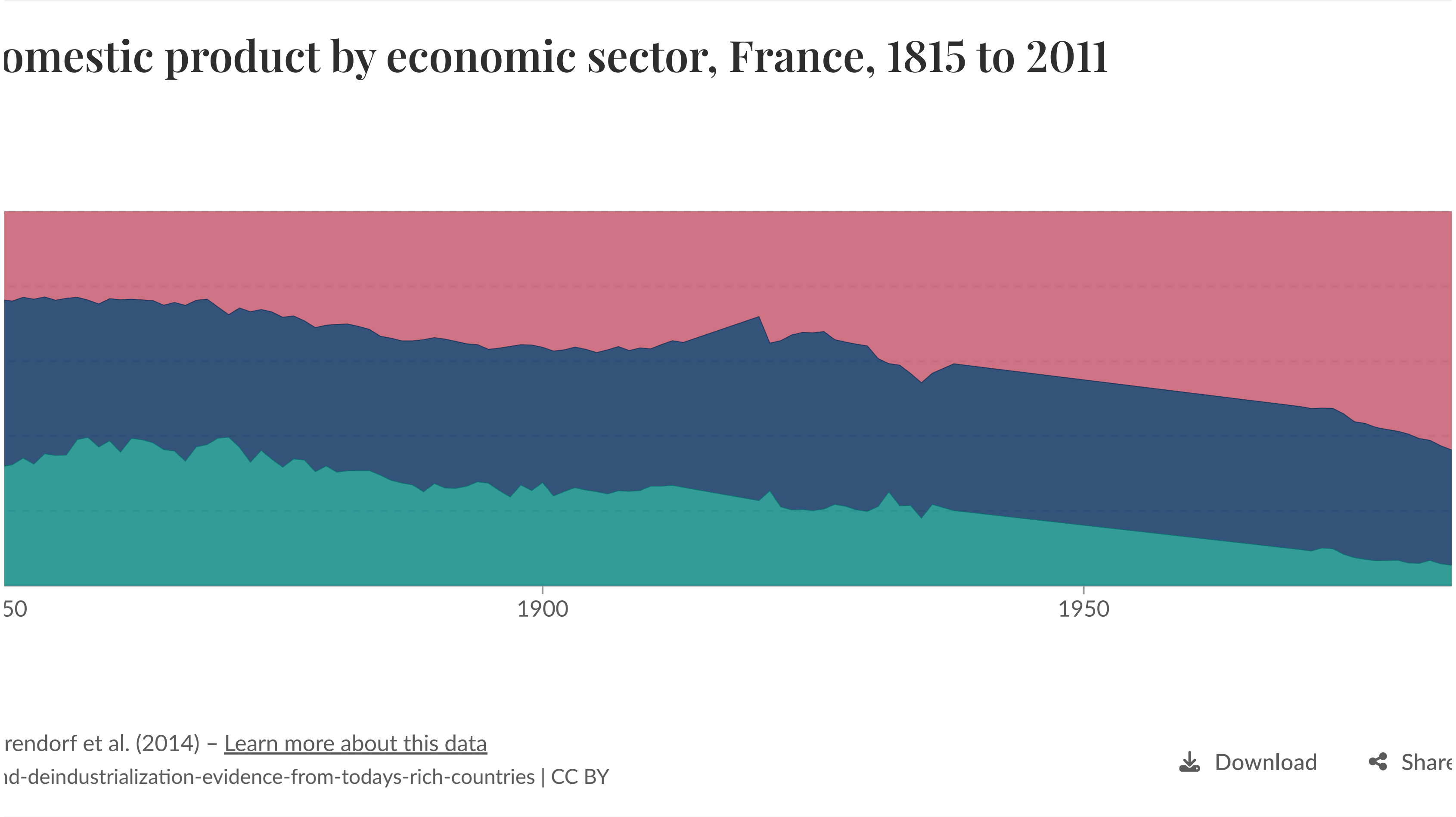
Structural transformation: From farm to firm

Capitalist firms harnessed technological innovation to boost productivity, reshaping the economy's structure.



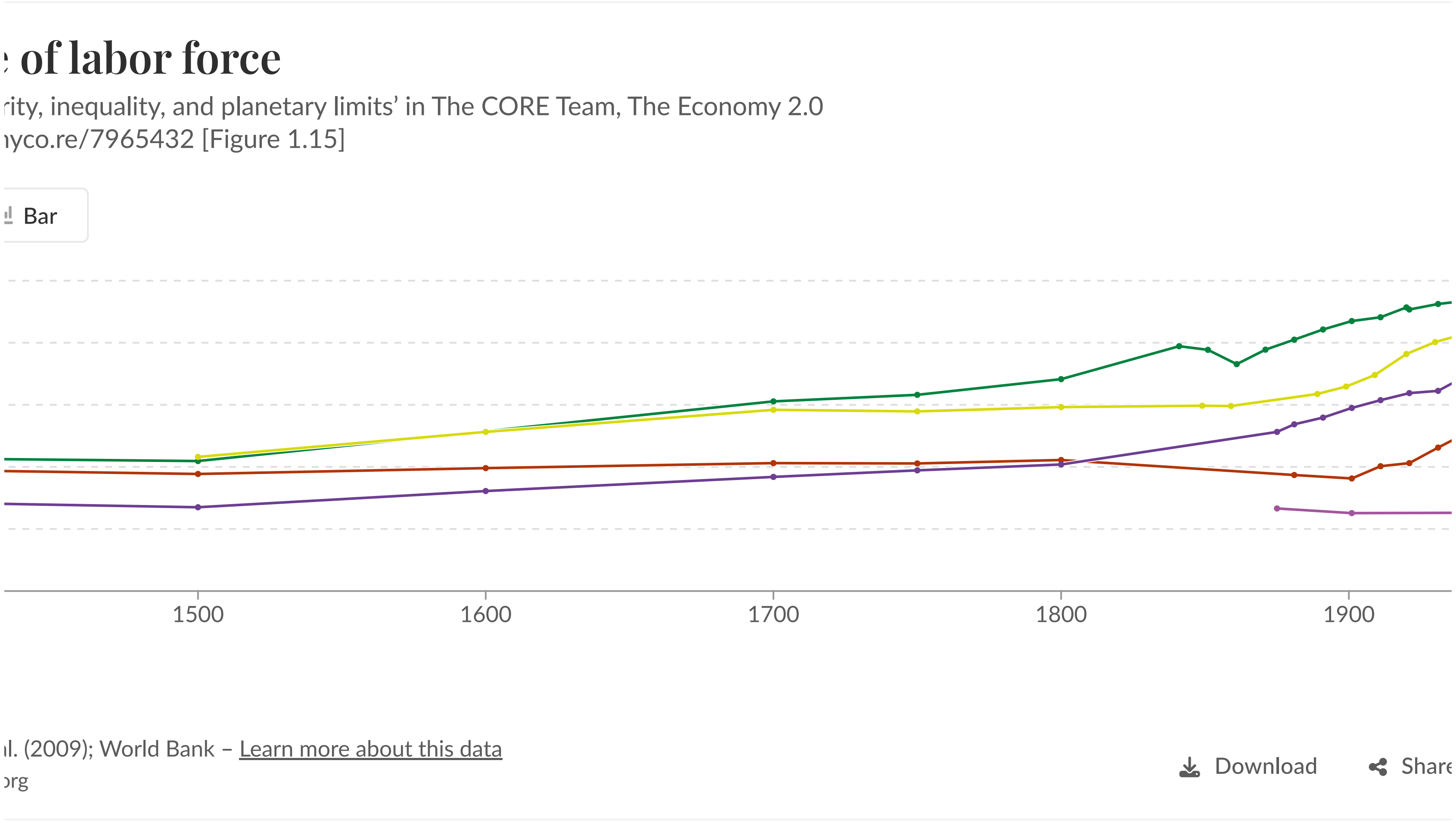
Structural transformation: From farm to firm

Capitalist firms harnessed technological innovation to boost productivity, reshaping the economy’s structure.



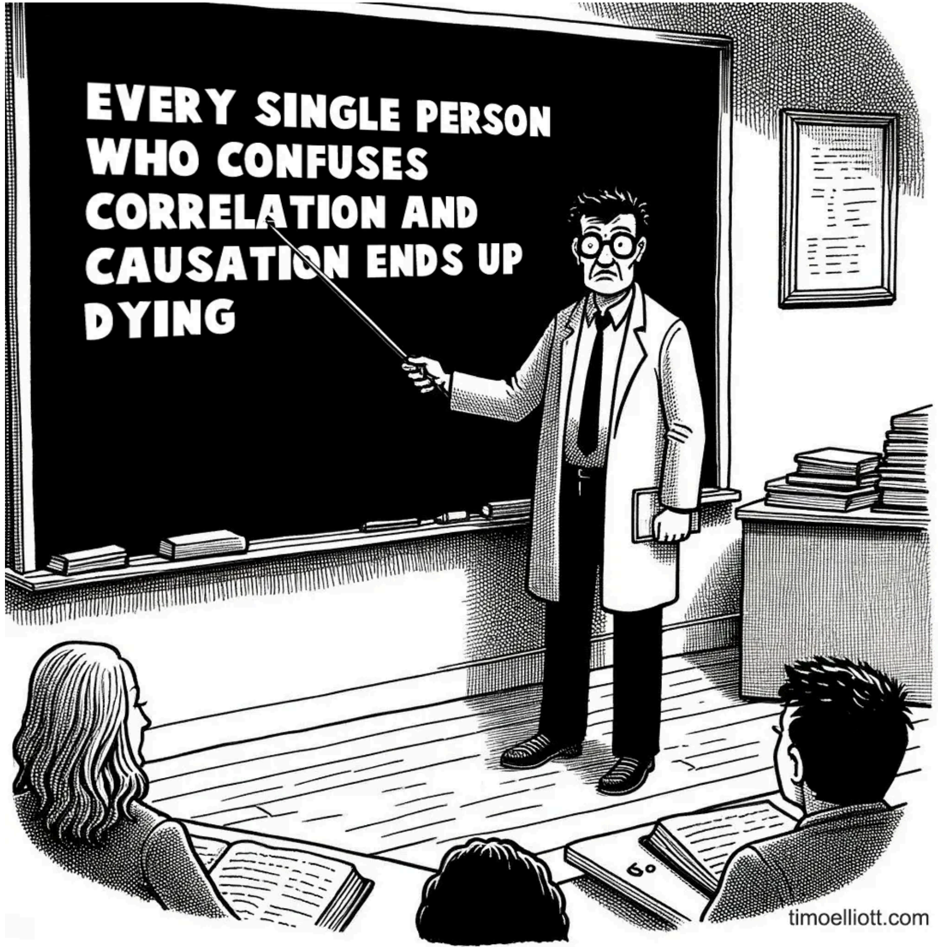
Structural transformation: From farm to firm

Capitalist firms harnessed technological innovation to boost productivity, reshaping the economy's structure.

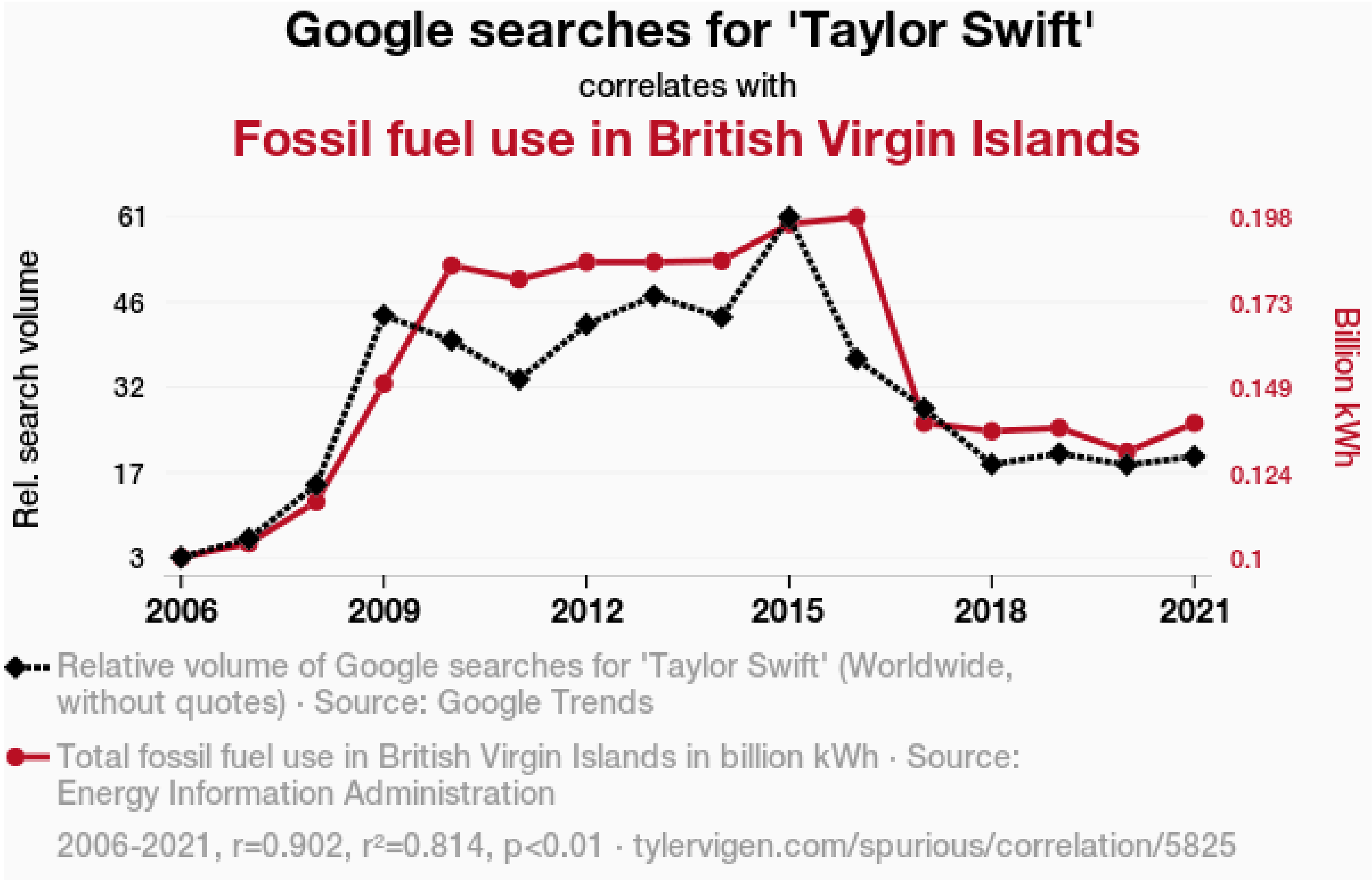


Capitalism, causation, and history's hockey stick

(Detour): Causation

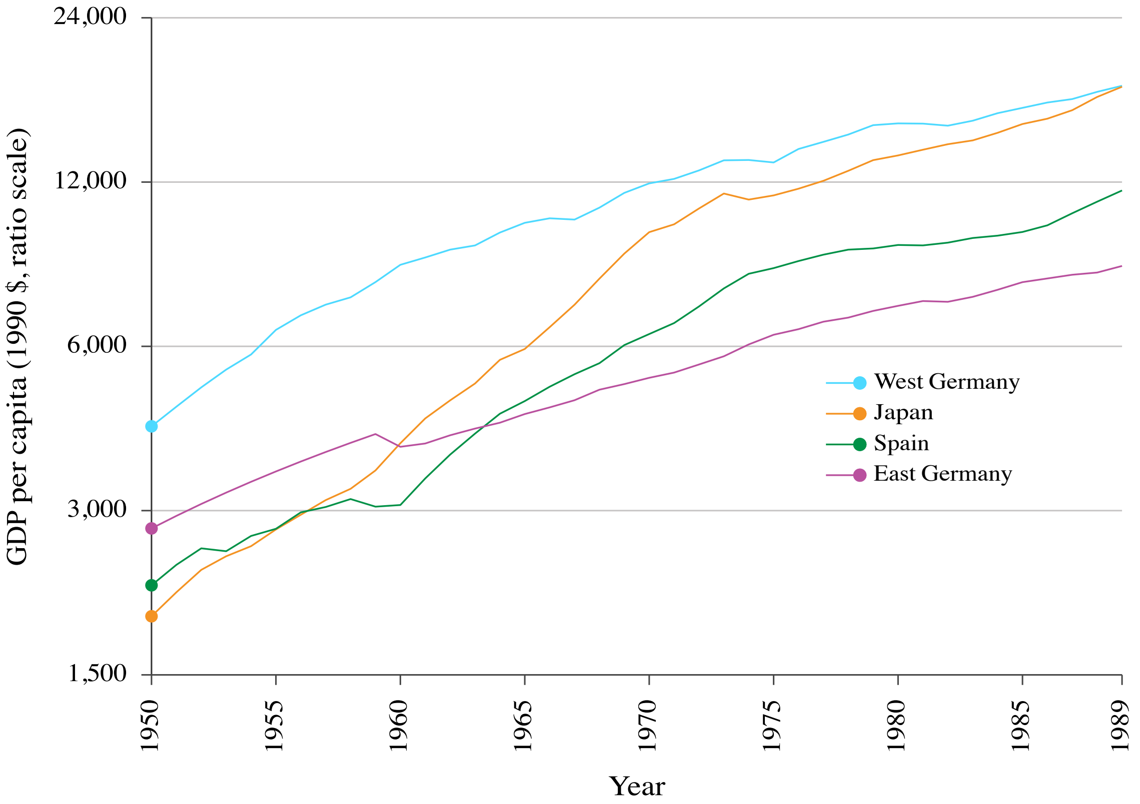


(Detour): Causation

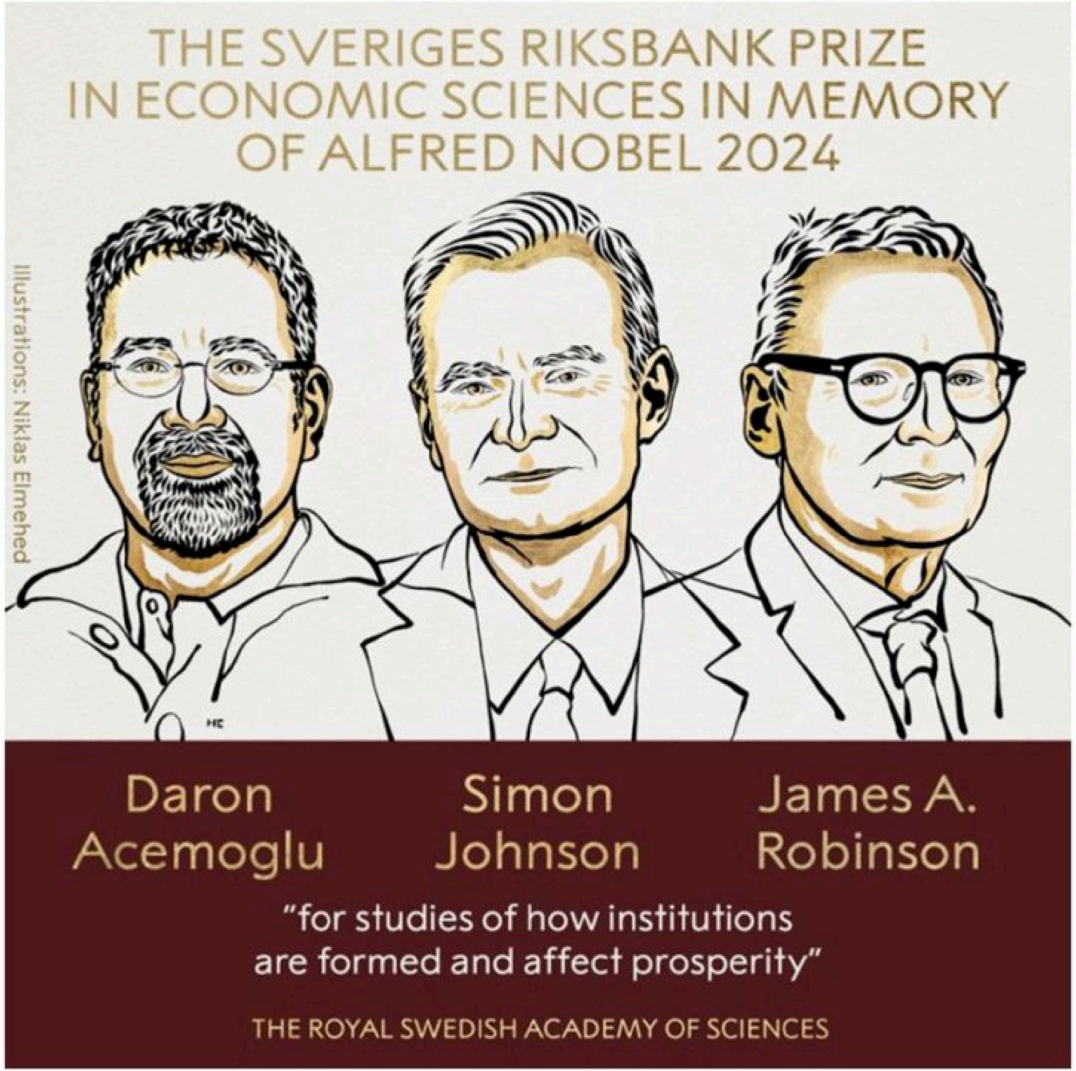


Did capitalism *cause* the hockey-stick growth?

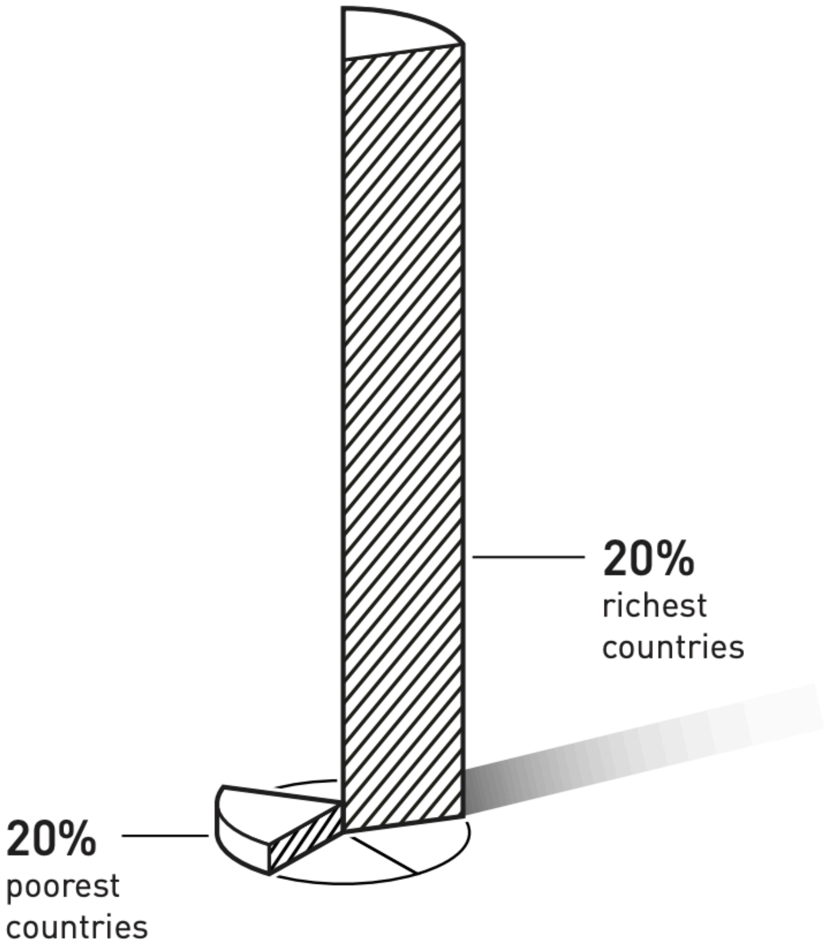
Natural experiment: the division of Germany at the end of World War II into two separate economic systems, capitalist in the west and centrally planned in the east.

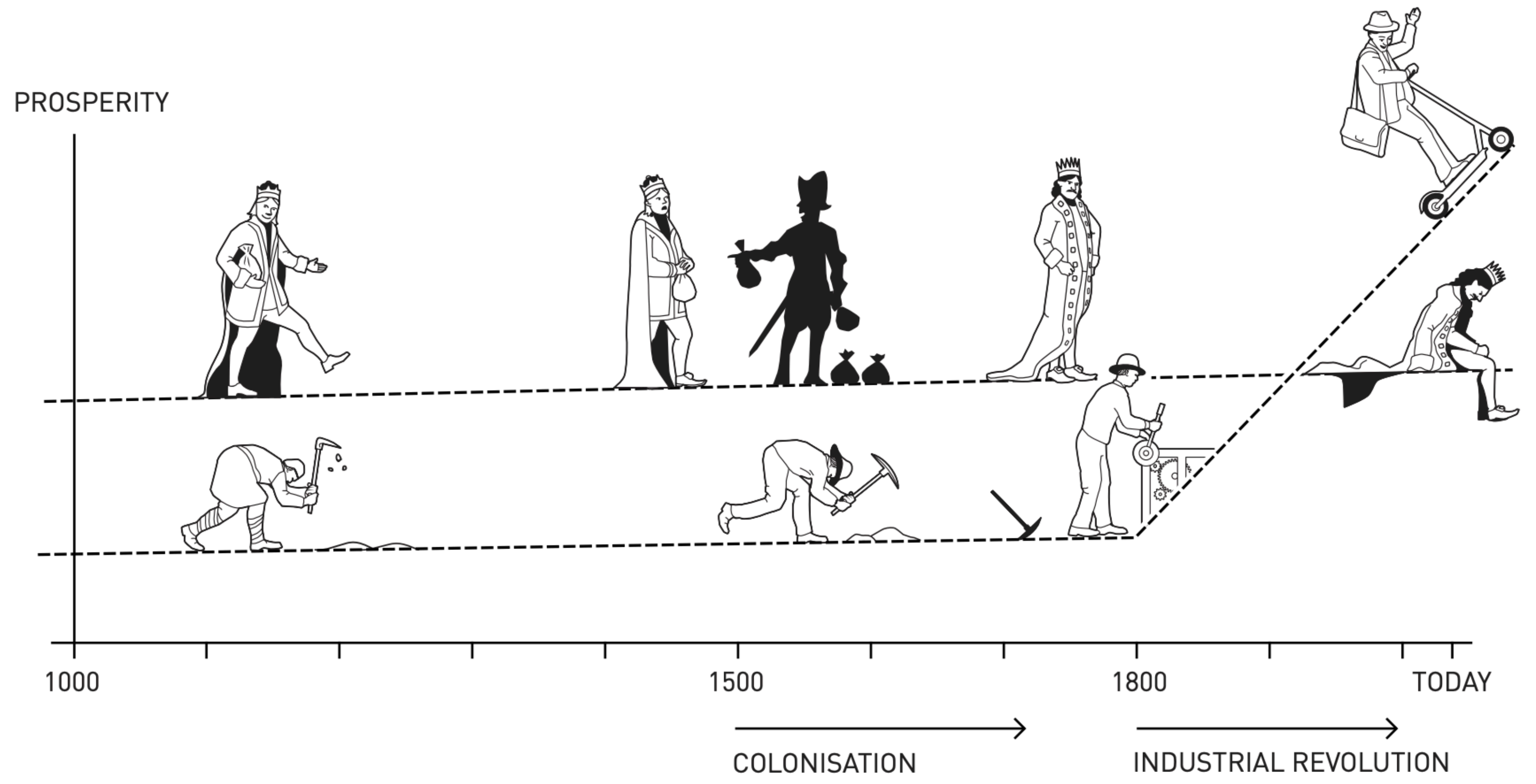


Why is there a persistent gap in GDP per capita or income between countries?


THE SVERIGES RIKSBANK PRIZE
IN ECONOMIC SCIENCES IN MEMORY
OF ALFRED NOBEL 2024
Illustrations: Niklas Elmehed
Daron
Acemoglu
Simon
Johnson
James A.
Robinson
"for studies of how institutions
are formed and affect prosperity"
THE ROYAL SWEDISH ACADEMY OF SCIENCES

GDP per capita



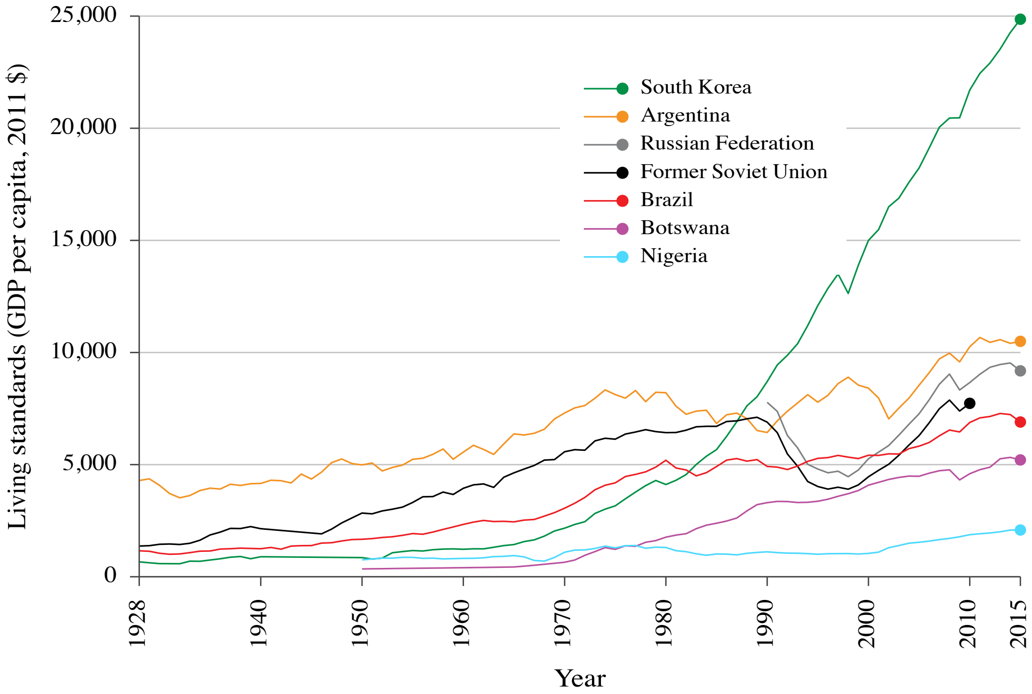

PROSPERITY
1000
1500
1800
TODAY
COLONISATION
INDUSTRIAL REVOLUTION

Varieties of capitalism: Institutions, government, and politics

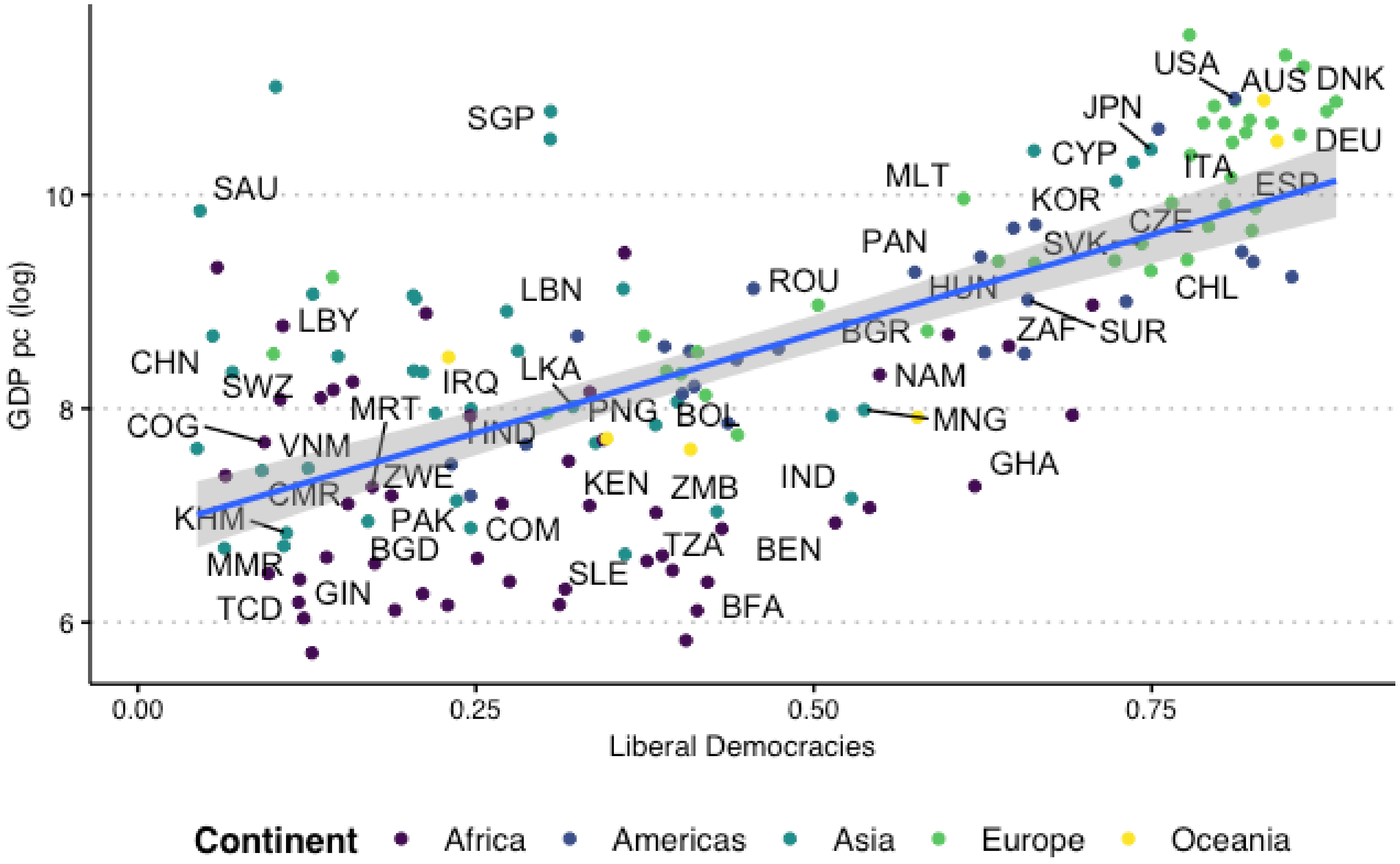
Divergence in growth

Not all capitalist economies are equally successful

- **economic conditions**: firms, private property, or markets may fail
- **political conditions**: capitalist institutions are regulated by the government
- the government also provides essential goods and services (infrastructure, education)



Democracy and Economic Development?



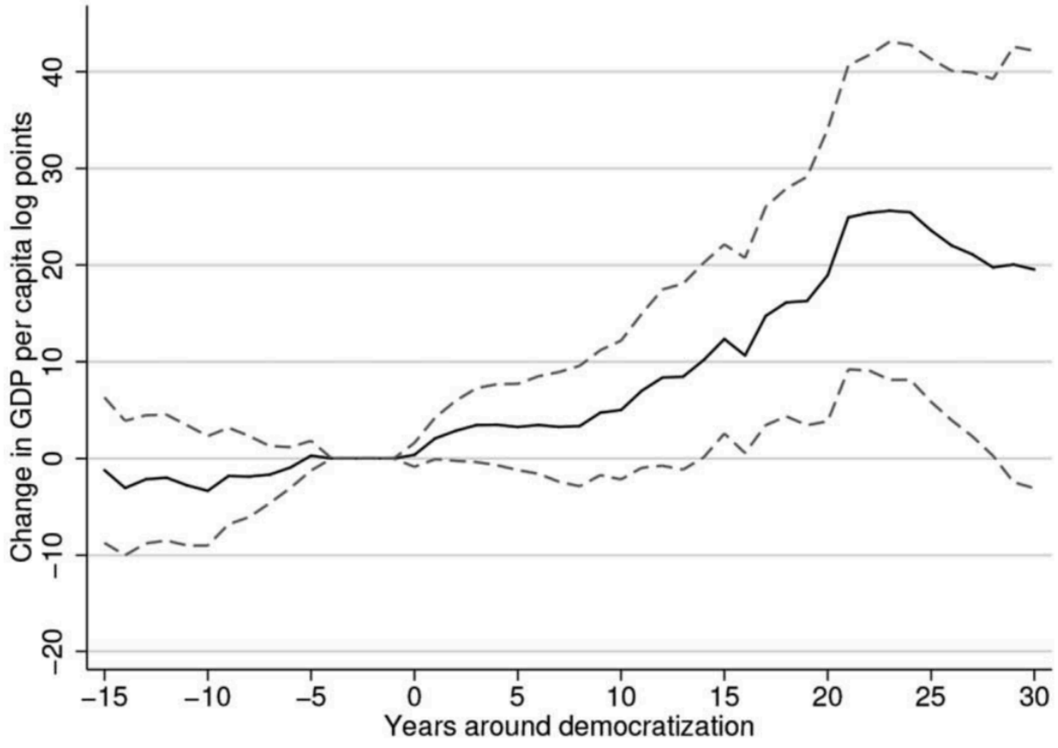


FIG. 3.—Semiparametric estimates of the over-time effects of democracy on the log of GDP, obtained with a regression model to estimate counterfactuals. This figure plots semi-parametric estimates of the effect of democratization on GDP per capita in log points. The solid line plots the estimated average effect on GDP per capita on countries that democratized (in log points), with a 95 percent confidence interval in dashed lines. Time (in years) relative to the year of democratization runs on the horizontal axis. The estimates are obtained by assuming and estimating a linear model for counterfactual outcomes, which we use to control for the influence of GDP dynamics. Section IV explains our approach in full detail.

Democracy Does Cause Growth (Acemoglu, Naidu, Restrepo, and Robinson, 2019)

Political systems

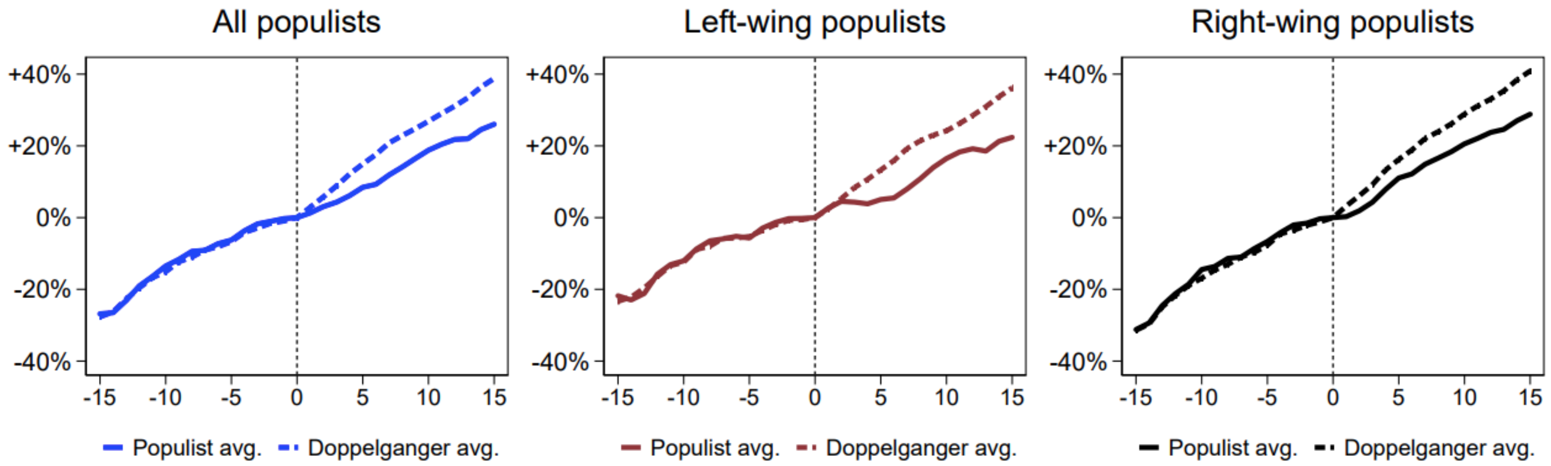
Capitalism coexists with many political systems.

A **political system** determines how governments will be selected, and how those governments will make and implement decisions.

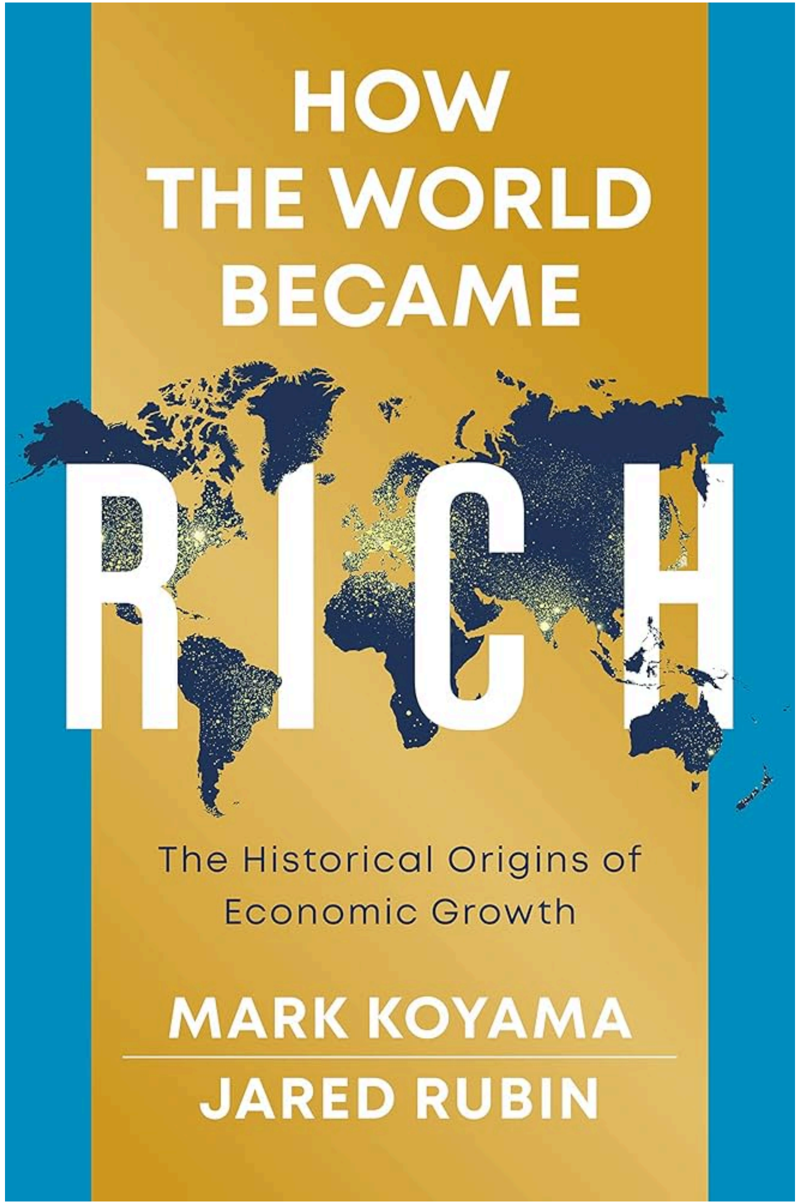
In most countries today, capitalism coexists with democracy

- individual rights of citizens (e.g. freedom of speech)
- fair elections

But capitalism has coexisted with non-democratic systems, too. Even democracy is 'not enough': populism.



Is there a silver bullet to explain these patterns?

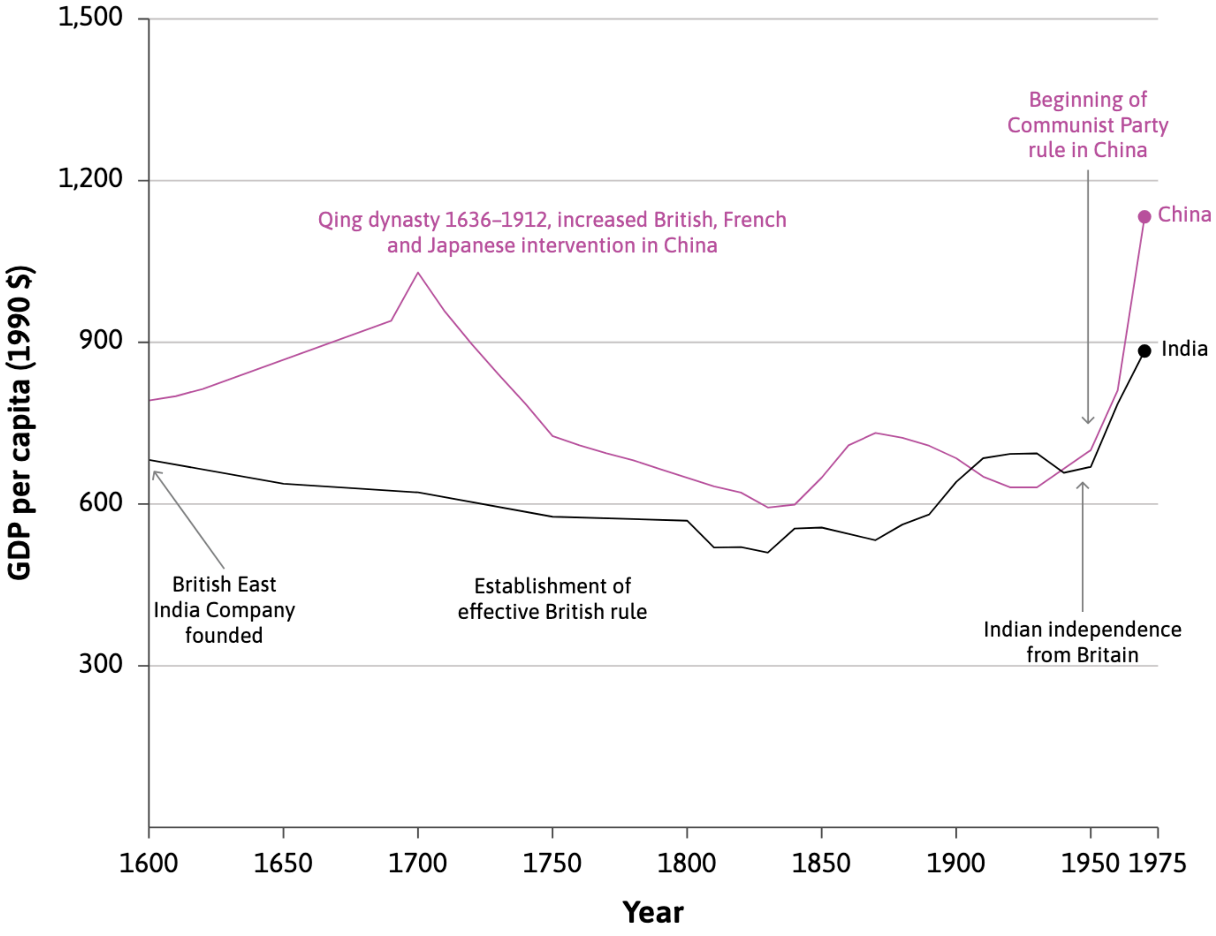


These are causal questions

- Geography
- Institutions
- Culture
- Demography
- Colonization and exploitation

1. Why did Europe got rich first?
2. Why did Britain got rich first?
3. Why the patterns expanded to other places?

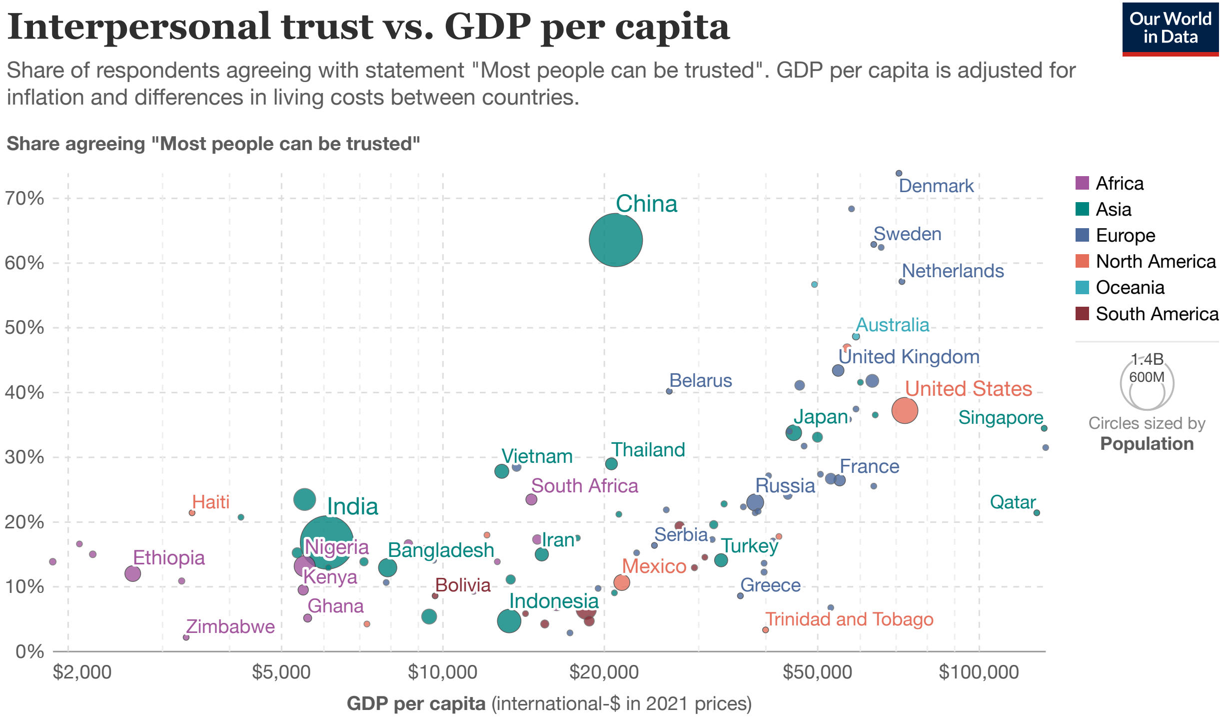
Colonization: Did the British colonization of India reduce Indian living standards?



Culture: Nexus between trust and economic development.

Interpersonal trust vs. GDP per capita

Share of respondents agreeing with statement "Most people can be trusted". GDP per capita is adjusted for inflation and differences in living costs between countries.

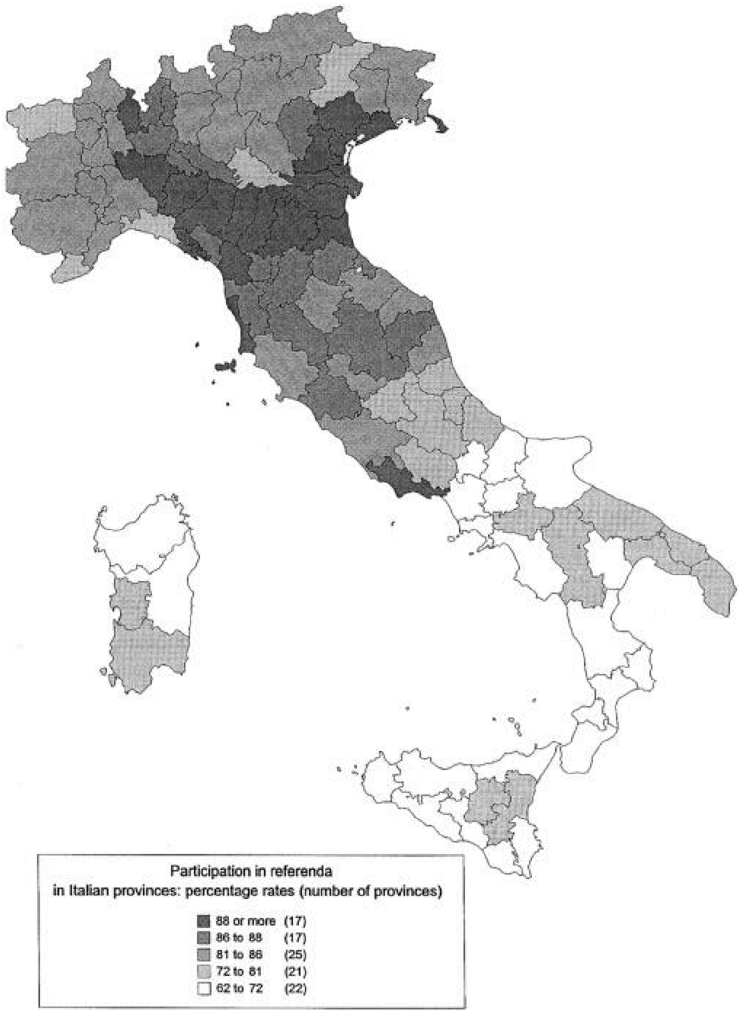


Data source: Integrated Values Surveys (2022); Data compiled from multiple sources by World Bank (2025)

Note: For each country, trust data is shown for the latest survey wave in the period 2009-2022. GDP per capita is expressed in international-$[1] at 2021 prices.

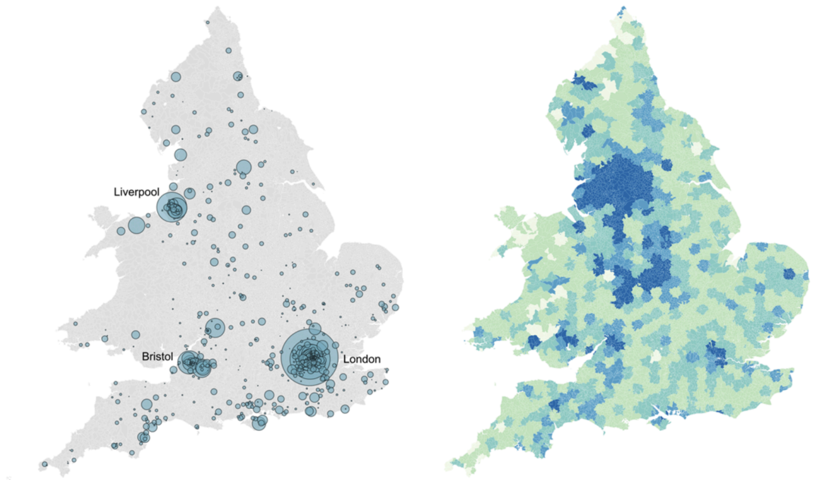
1. International dollars: International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: What are Purchasing Power Parity adjustments and why do we need them?

Culture: Italy as a case study

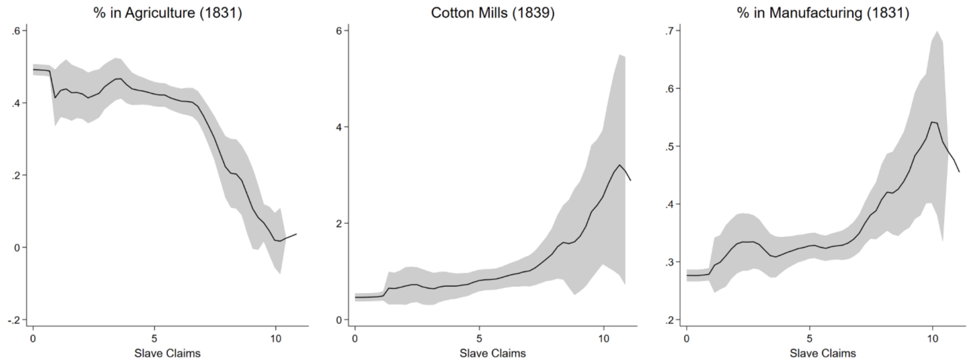


The Role of Social Capital in Financial Development (Guiso, Sapienza, and Zingales, 2004)

Colonization and Exploitation: Britain as a case study



(a) Slaveholder compensation in 1833 (b) Manufacturing employment share in 1831

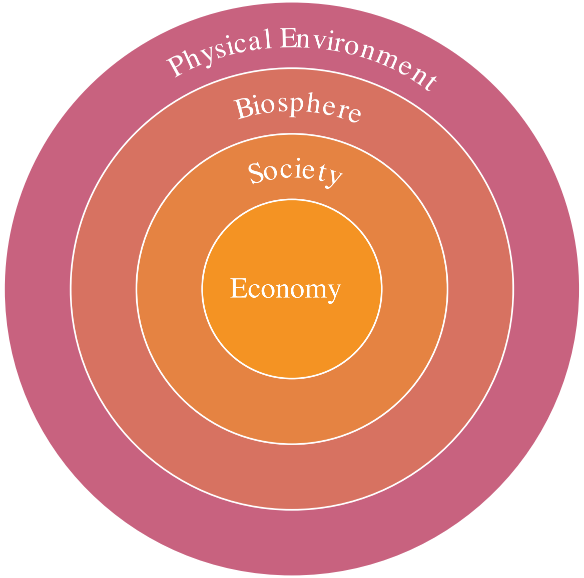


Slavery and the British Industrial Revolution (Heblich, Redding, and Voth, 2023)

Economics, the economy, and the biosphere

Economics, the economy, and the biosphere

Economics: the study of **how people interact with each other** and **with their natural environment** in producing and acquiring their livelihoods, and how this changes over time and differs across societies.



How human and ecosystem health are intertwined: Evidence from vulture population collapse in India (Frank and

Summary

Summing up

1) Important trends in economic variables over time

- Income inequality across regions has increased a lot over time
- “Hockey-stick” growth in GDP and its environmental consequences
- Technological progress helped bring about these trends

2) The adoption of capitalism was another key factor

- Capitalism = Private property + Markets + Firms
- Failure of these institutions can explain divergence in economic growth across countries
- Political systems and the role of government also determine the type of capitalist society

Next class

- Using economic models to explain the trends in technological growth over time
- The role of firms in technological development